

Chapter 18

Application of Derivatives

Solutions

SECTION - A

Objective Type Questions (One option is correct)

[Rate of Change of Quantities]

1. The radius of a right circular cylinder increases at the rate of 0.1 cm/minute, and the height decreases at the rate of 0.2 cm/minute. The rate of change of the volume of the cylinder, in $\text{cm}^3/\text{minute}$, when the radius is 2 cm and the height is 3 cm is

- (1) -2π (2) $-\frac{8\pi}{5}$ (3) $-\frac{3\pi}{5}$ (4) $\frac{2\pi}{5}$

Sol. Answer (4)

$$V = \pi r^2 h$$

$$dV = 2\pi r h \, dr + \pi r^2 \, dh$$

$$\Rightarrow dV = 2\pi \times 2 \times 3 \times (0.1) + \pi \times 2^2 \times (-0.2)$$

$$= 1.2\pi - 0.8\pi$$

$$= 0.4\pi = \frac{2\pi}{5}$$

2. A point on the parabola $y^2 = 18x$ at which the ordinate increases at twice the rate of abscissa is

- (1) (2, 4) (2) (2, -4) (3) $\left(-\frac{9}{8}, \frac{9}{2}\right)$ (4) $\left(\frac{9}{8}, \frac{9}{2}\right)$

Sol. Answer (4)

$$y^2 = 18x$$

$$\Rightarrow 2y \frac{dy}{dt} = 18 \frac{dx}{dt}$$

$$\Rightarrow 2y \frac{2dx}{dt} = 18 \frac{dx}{dt}$$

$$4y = 18$$

$$\Rightarrow y = \frac{9}{2}$$

$$\frac{81}{4} = 18x$$

$$\Rightarrow x = \frac{9}{8}$$

3. Coffee is draining from a conical filter of height and diameter both 15 cm into a cylindrical coffee pot of diameter 15 cm. The rate at which coffee drains from the filter into the pot is $100 \text{ cm}^3/\text{minute}$. The rate in cm/minute at which the level in the pot is rising at the instant when the coffee in the pot is 10 cm, is

(1) $\frac{9}{16\pi}$

(2) $\frac{25}{9\pi}$

(3) $\frac{5}{3\pi}$

(4) $\frac{16}{9\pi}$

Sol. Answer (4)

Rate of change of volume in the cylinder = Rate of change of volume in conical filter.

$$\pi r^2 h = V$$

Differentiate w.r.t. t

$$\frac{dV}{dt} = \pi r^2 \frac{dh}{dt}$$

$$\Rightarrow 100 = \pi \times \left(\frac{15}{2}\right)^2 \frac{dh}{dt}$$

$$\Rightarrow \frac{dh}{dt} = \frac{16}{9\pi} \text{ cm/s}$$

4. The radius of a right circular cylinder increases at a constant rate. Its altitude is a linear function of the radius and increases three times as fast as radius. When the radius is 1 cm the altitude is 6 cm. When the radius is 6 cm, the volume is increasing at the rate of $1 \text{ cm}^3/\text{s}$. When the radius is 36 cm, the volume is increasing at a rate of $n \text{ cm}^3/\text{s}$. The value of ' n ' is equal to

(1) 12

(2) 22

(3) 30

(4) 33

Sol. Answer (4)

$$\frac{dr}{dt} = k \quad \frac{dh}{dt} = 3k$$

$$h = \alpha r + c$$

$$\frac{dh}{dt} = \alpha \frac{dr}{dt}$$

$$\alpha = 3$$

$$h = 3r + c$$

$$h = 6, r = 1$$

$$c = 3$$

$$\text{Volume} = \pi r^2 (3r + 3)$$

$$\frac{dV}{dt} = 3\pi[3r^2 + 2r] \frac{dr}{dt}$$

$$\Rightarrow 1 = 3\pi[108 + 12] \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1}{360\pi}$$

$$\begin{aligned}\frac{dV}{dt} &= 3\pi r(3r+2)\frac{dr}{dt} \\ &= 3\pi \times 36(110) \times \frac{1}{40\pi \times 9} \\ &= 33 \text{ cubic cm/second.}\end{aligned}$$

5. A particle moves along the curve $y = x^{\frac{3}{2}}$ in the first quadrant in such a way that its distance from the origin increases at the rate of 11 units per second. The value of $\frac{dx}{dt}$ where $x = 3$ is, t is time in seconds

- (1) 4 (2) $\frac{9}{2}$ (3) $\frac{3\sqrt{3}}{2}$ (4) 12

Sol. Answer (1)

$$y = x^{\frac{3}{2}}$$

$$\Rightarrow y^2 = x^3$$

Distance from origin

$$D = \sqrt{x^2 + y^2}$$

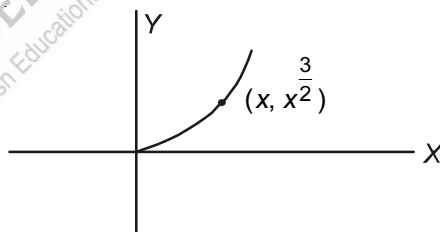
Squaring both side and differentiating

$$2DD' = (2x + 3x^2)\frac{dx}{dt}$$

$$\Rightarrow 2D\frac{dD}{dt} = \frac{2x + 3x^2}{2x\sqrt{1+x}}\frac{dx}{dt}$$

$$\Rightarrow 11 = \frac{2+3x}{2\sqrt{1+x}}\frac{dx}{dt}$$

$$\left(\frac{dx}{dt}\right)_{x=3} = \frac{11 \times 2 \times 2}{11} = 4$$



6. A cube of ice melts without changing shape at the uniform rate of 4 cm³/minute. The rate of change of the surface area of the cube, in cm²/minute, when the volume of the cube is 125 cm³, is

- (1) -4 (2) $-\frac{16}{5}$ (3) $-\frac{16}{6}$ (4) $-\frac{8}{15}$

Sol. Answer (2)

$$V = a^3$$

$$\Rightarrow \frac{dV}{dt} = 3a^2 \frac{da}{dt}$$

$$\Rightarrow 4 \text{ cm}^3/\text{minute} = 3 \times 25 \frac{da}{dt}$$

$$\Rightarrow S = 6a^2$$

$$\Rightarrow \frac{dS}{dt} = -12a \frac{da}{dt}$$

$$\Rightarrow \frac{dS}{dt} = -12 \times 5 \times \frac{4}{75} = -\frac{16}{5}$$

7. Two cyclists start from the junction of two perpendicular roads, their velocities being $3v$ metres/minute and $4v$ metres/minute. The rate at which the two cyclists are separating is

- (1) $\frac{7}{2}v$ m/min (2) $5v$ m/min (3) v m/min (4) $7v$ m/min

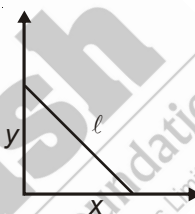
Sol. Answer (2)

$$\ell^2 = x^2 + y^2$$

$$\frac{dx}{dt} = 3v \text{ m/minute}, \frac{dy}{dt} = 4v \text{ m/minute}$$

$$\Rightarrow 2\ell \frac{d\ell}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt} \Rightarrow \ell \frac{d\ell}{dt} = x3v + y4v$$

$$\Rightarrow \frac{d\ell}{dt} = \frac{x}{\ell} 3v + \frac{y}{\ell} 4v = \frac{3}{5} 3v + \frac{4}{5} 4v = 5v$$



[Monotonicity]

8. The interval on which $f(x) = 2x^3 + 9x^2 + 12x - 1$ is decreasing in

- (1) $(-1, \infty)$ (2) $(-2, -1)$ (3) $(-\infty, -2)$ (4) $(-1, 1)$

Sol. Answer (2)

$$f(x) = 2x^3 + 9x^2 + 12x - 1$$

$$f'(x) = 6x^2 + 18x + 12$$

For decreasing

$$f'(x) < 0$$

$$6x^2 + 18x + 12 < 0$$

$$x^2 + 3x + 2 < 0$$

$$(x + 1)(x + 2) < 0$$

$$x \in (-2, -1)$$

9. The exhaustive values of x for which $f(x) = \cot^{-1} x + x$ is increasing is

- (1) $(1, \infty)$ (2) $(-1, \infty)$ (3) $(-\infty, \infty)$ (4) $(0, \infty)$

Sol. Answer (3)

$$f(x) = \cot^{-1} x + x$$

$$f'(x) = \frac{-1}{1+x^2} + 1 = \frac{+x^2}{1+x^2}$$

$$f'(x) > 0$$

$$x \in R$$

10. The function $f(x) = x^3 - 3x$ is(1) Increasing in $(-\infty, -1) \cup (1, \infty)$ and decreasing in $(-1, 1)$ (2) Decreasing in $(-\infty, -1) \cup (1, \infty)$ and increasing in $(-1, 1)$ (3) Increasing in $(0, \infty)$ and decreasing in $(-\infty, 0)$ (4) Decreasing in $(0, \infty)$ and increasing in $(-\infty, 0)$ **Sol.** Answer (1)

$$f(x) = x^3 - 3x$$

$$f'(x) = 3x^2 - 3$$

$$= 3(x^2 - 1)$$

$$f'(x) = 3(x+1)(x-1)$$

For increasing function

$$f'(x) > 0$$

$$x \in (-\infty, -1) \cup (1, \infty)$$

11. A function $y = f(x)$ is given by $x = \frac{1}{1+t^2}$ and $y = \frac{1}{t(1+t^2)}$ for all $t > 0$, then f is(1) Increasing for $t \in \left(0, \frac{3}{2}\right)$ and decreasing for $t \in \left(\frac{3}{2}, \infty\right)$ (2) Decreasing for $t \in \left(0, \frac{1}{2}\right)$ (3) Increasing for $t \in (0, \infty)$ (4) Decreasing for $t \in (0, 1)$ **Sol.** Answer (3)

$$x = \frac{1}{1+t^2}, \quad y = \frac{1}{t(1+t^2)}$$

$$\frac{dx}{dt} = \frac{-2t}{(1+t^2)^2}, \quad \frac{dy}{dt} = \frac{-(1+3t^2)}{t^2(1+t^2)^2}$$

$$\frac{dy}{dx} = \frac{(1+3t^2)}{t^2 \times 2t}$$

$$\frac{dy}{dx} = \frac{1+3t^2}{2t^3}$$

$$\frac{dy}{dx} > 0, \text{ if } t > 0$$

12. If $f(x) = x^3 + 4x^2 + \lambda x + 1$ is a monotonically decreasing function of x in the largest possible interval $\left(-2, -\frac{2}{3}\right)$

then

(1) $\lambda = 4$

(2) $\lambda = 2$

(3) $\lambda = -1$

(4) λ has no real value

Sol. Answer (1)

$$f(x) = x^3 + 4x^2 + \lambda x + 1$$

$$\Rightarrow f'(x) = 3x^2 + 8x + \lambda$$

For monotonic decreasing $f'(x) < 0$

$$\Rightarrow 3x^2 + 8x + \lambda < 0 \quad \Rightarrow \frac{-8 - \sqrt{64 - 12\lambda}}{6} < x < \frac{-8 + \sqrt{64 - 12\lambda}}{6}$$

According to question $-2 < x < -\frac{2}{3}$

Hence $-2 \leq \frac{-8 - \sqrt{64 - 12\lambda}}{6}$ and $-\frac{2}{3} \leq \frac{-8 + \sqrt{64 - 12\lambda}}{6}$

$$\Rightarrow \lambda \geq 4 \text{ and } \lambda \leq 4 \quad \Rightarrow \lambda = 4$$

13. If a function $f(x)$ increases in the interval (a, b) then the function $\phi(x) = [f(x)]^n$ increases in the same interval and $\phi(x) \neq f(x)$ if

(1) $n = -1$

(2) $n = 0$

(3) $n = 3$

(4) $n = 4$

Sol. Answer (3)

$$\phi(x) = [f(x)]^n \quad \Rightarrow \phi'(x) = n[f(x)]^{n-1} \cdot f'(x)$$

as $f(x)$ is increasing function in (a, b) hence $f'(x) > 0, \forall x \in (a, b)$

if $\phi f(x)$ is increasing function then

$$n - 1 = 2, n = 3.$$

14. The function f , defined by $f(x) = \frac{x^2}{2} + \ln x - 2\cos x$ increases for $x \in$

(1) R^-

(2) R^+

(3) $R - \{0\}$

(4) $[1, \infty)$

Sol. Answer (2)

$$f(x) = \frac{x^2}{2} + \ln x - 2\cos x \Rightarrow f'(x) = x + \frac{1}{x} + 2\sin x$$

We know that $x + \frac{1}{x} \geq 2, \forall x > 0$ and $2\sin x \leq 2, \forall x \in R$

$\Rightarrow f'(x) > 0$ for $x > 0$

Hence $f(x)$ is increasing in $(0, \infty)$

When $x < 0, x + \frac{1}{x} \leq -2$ but $-2 \leq 2\sin x \leq 2, \forall x \in R$

$\Rightarrow f'(x) < 0$ hence $f(x)$ is decreasing in $(-\infty, 0)$

15. Let $f(x) = xe^{x(1-x)}$, then $f(x)$ is

(1) Increasing on R

(2) Increasing on $\left[-\frac{1}{2}, 1\right]$

(3) Decreasing on R

(4) Decreasing on $\left[-\frac{1}{2}, 1\right]$

Sol. Answer (2)

$$f(x) = xe^{x(1-x)}$$

$$\Rightarrow f'(x) = e^{x(1-x)}(1+x-2x^2)$$

$$\Rightarrow f'(x) = -(x-1)(2x+1)e^{x(1-x)}$$

Clearly $f(x)$ is increasing on $\left[-\frac{1}{2}, 1\right]$

16. Which of the following is correct?

(1) $\ln(1+x) < x \quad \forall -1 < x \leq 0$

(2) $\ln(1+x) < x \quad \forall 0 < x$

(3) $\ln(1+x) > x \quad \forall x > 0$

(4) $\ln(1+x) < x \quad \forall x > -1$

Sol. Answer (2)

We know that

If $x > 0$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty \Rightarrow e^x > 1 + x \Rightarrow x > \ln(1+x)$$

Second Method

Let $f(x) = x - \ln(1 + x)$

$$\Rightarrow f'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x}$$

$$\Rightarrow f'(x) > 0 \text{ for } x \in (-\infty, -1) \cup (0, \infty) \text{ and } f'(x) < 0 \text{ for } x \in (-1, 0)$$

$$\Rightarrow f(x) > f(0) \text{ for } x \in (0, \infty)$$

$$\Rightarrow x - \ln(1 + x) > 0$$

$$\Rightarrow x > \ln(1 + x), \forall x > 0$$

17. $f(x) = \frac{x}{\sin x}$ and $g(x) = \frac{x}{\tan x}$, where $0 < x \leq 1$, then in the interval

(1) Both $f(x)$ and $g(x)$ are increasing functions

(2) Both $f(x)$ and $g(x)$ are decreasing functions

(3) $f(x)$ is an increasing function

(4) $g(x)$ is an increasing function

Sol. Answer (3)

$$f(x) = \frac{x}{\sin x} \Rightarrow f'(x) = \frac{\sin x - x \cos x}{\sin^2 x}$$

Let $\sin x - x \cos x = h(x)$

$$\Rightarrow h'(x) = \cos x - [\cos x - x \sin x] = x \sin x \Rightarrow h'(x) > 0 \forall x \in (0, 1)$$

$$\Rightarrow f'(x) > 0 \text{ i.e., } f(x) \text{ is increasing function in } (0, 1]$$

$$\text{Now } g(x) = \frac{x}{\tan x} \Rightarrow g'(x) = \frac{\tan x - x \sec^2 x}{\tan^2 x}$$

Let $\tan x - x \sec^2 x = \phi(x)$

$$\phi'(x) = \sec^2 x - [\sec^2 x + 2x \sec^2 x \tan x] = -2x \sec x \tan x$$

$$\phi'(x) < 0, \forall x \in (0, 1]$$

$$\Rightarrow g(x) \text{ is decreasing in } (0, 1]$$

18. If $f(x) = \frac{P^2 - 1}{P^2 + 1} x^3 - 3x + \log 2$ is a decreasing function of x in R then the set of possible values of P (independent of x) is

(1) $[-1, 1]$

(2) $(1, \infty)$

(3) $(-\infty, 1]$

(4) $(2, \infty)$

Sol. Answer (1)

$$f(x) = \frac{P^2 - 1}{P^2 + 1} x^3 - 3x + \log 2$$

$$f'(x) = \left(\frac{p^2 - 1}{p^2 + 1} \right) 3x^2 - 3$$

For decreasing function

$$f'(x) < 0$$

$$\left(\frac{p^2 - 1}{p^2 + 1} \right) x^2 - 1 < 0 \Rightarrow \left(\frac{p^2 - 1}{p^2 + 1} \right) \leq 0$$

$$\Rightarrow -1 \leq p \leq 1 \Rightarrow p \in [-1, 1]$$

19. Let $f(x) = (x - p)^2 + (x - q)^2 + (x - r)^2$. Then $f(x)$ has a minimum at $x = \lambda$, where λ is equal to

(1) $\frac{p+q+r}{3}$

(2) $\sqrt[3]{pqr}$

(3) $\frac{3}{\frac{1}{p} + \frac{1}{q} + \frac{1}{r}}$

(4) $\frac{3}{p+q+r}$

Sol. Answer (1)

$$f(x) = (x - p)^2 + (x - q)^2 + (x - r)^2 = 3x^2 - 2x(p + q + r) + (p^2 + q^2 + r^2)$$

We know that $f(x) = ax^2 + bx + c$ (if $a > 0$)

$$\text{is minimum at } x = -\frac{b}{2a}$$

$$\text{Hence } \lambda = \frac{p+q+r}{3}$$

Second method :

$$\text{For minimum value of } f(x), \frac{df(x)}{dx} = 0 \text{ and } \frac{d^2f(x)}{dx^2} > 0$$

$$\text{We observe that } \frac{df(x)}{dx} = 6x - 2(p + q + r) = 0$$

$$x = \frac{p+q+r}{3}$$

$$\frac{d^2f(x)}{dx^2} = 6 = +ve$$

$$\text{Hence } f(x) \text{ is minimum at } x = \frac{p+q+r}{3}$$

20. Let $f''(x) > 0 \forall x \in R$ and let $g(x) = f(x) + f(2 - x)$ then interval of x for which $g(x)$ is increasing is

(1) $(1, \infty)$

(2) R

(3) $[-1, 1]$

(4) $[-2, \infty)$

Sol. Answer (1)

$$g(x) = f(x) + f(2 - x)$$

$$g'(x) = f'(x) - f'(2-x)$$

for g to be increase

$$x > 2 - x$$

$$x > 1$$

21. If θ is a positive acute angle then

$$(1) \tan \theta < \theta < \sin \theta$$

$$(2) \theta < \sin \theta < \tan \theta$$

$$(3) \sin \theta < \tan \theta < \theta$$

$$(4) \sin \theta < \theta < \tan \theta$$

Sol. Answer (4)

$$\text{Let } f(\theta) = \theta - \sin \theta$$

$$\Rightarrow f'(\theta) = 1 - \cos \theta$$

$$\Rightarrow f'(\theta) > 0 \quad \forall \theta \in \left(0, \frac{\pi}{2}\right) \text{ which shows that } f(\theta) \text{ is increasing in } \left(0, \frac{\pi}{2}\right)$$

$$\text{Hence } f(\theta) > f(0)$$

$$\Rightarrow \theta - \sin \theta > 0$$

$$\theta > \sin \theta$$

...(1)

$$\text{Again let } g(\theta) = \tan \theta - \theta$$

$$\Rightarrow g'(\theta) = \sec^2 \theta - 1$$

$$\Rightarrow g'(\theta) > 0, \quad \forall \theta \in \left(0, \frac{\pi}{2}\right) \text{ which shows that } g(\theta) \text{ is increasing in } \left(0, \frac{\pi}{2}\right)$$

$$g(\theta) > g(0)$$

$$\Rightarrow \tan \theta - \theta > 0$$

$$\Rightarrow \tan \theta > \theta$$

...(2)

Combining (1) and (2)

$$\tan \theta > \theta > \sin \theta$$

22. If the equation $x^3 + px^2 + qx + 1 = 0$ ($p < q$) has only one real root α , then α belongs to

$$(1) (-2, -1)$$

$$(2) (-1, 0)$$

$$(3) (0, 1)$$

$$(4) (1, 2)$$

Sol. Answer (2)

$$x^3 + px^2 + qx + 1 = 0 \quad (p < q)$$

$$\text{Let } f(x) = x^3 + px^2 + qx + 1$$

$$f(0) = 1 \Rightarrow f(0) > 0$$

$$f(-1) = p - q \Rightarrow f(-1) < 0 \quad \Rightarrow f(0) \cdot f(-1) < 0$$

$$\Rightarrow f(x) = 0 \text{ has one real root in } (-1, 0) \quad \Rightarrow \alpha \in (-1, 0)$$

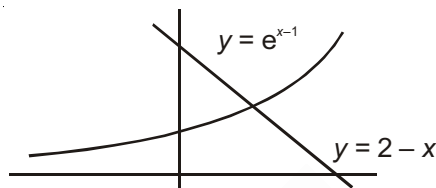
23. The equation $e^{x-1} + x - 2 = 0$ has

- (1) One real root (2) Two real roots
(3) Three real roots (4) Four real roots

Sol. Answer (1)

The given equation can be put in the form $e^{x-1} = 2 - x$

Let us draw the graphs of $y = e^{x-1}$ and $y = 2 - x$ and the point of intersection of the two curves gives the number of real roots of the equation.



From the graph of the two curves it follows that the given equation has one real root.

24. If $0 < \alpha < \beta < \gamma < \frac{\pi}{2}$ and

$$x = \frac{\sin \alpha + \sin \beta + \sin \gamma}{\cos \alpha + \cos \beta + \cos \gamma} \text{ then}$$

- (1) $x \in [\tan \alpha, \tan \gamma]$ (2) $x \in (\tan \alpha, \tan \gamma)$
(3) $x \in (\cot \alpha, \cot \gamma)$ (4) $x \in [\cot \alpha, \cot \gamma]$

Sol. Answer (2)

$$0 < \alpha < \beta < \gamma < \frac{\pi}{2}$$

$$\Rightarrow \sin \alpha < \sin \beta < \sin \gamma$$

$$\Rightarrow 3 \sin \alpha < \sin \alpha + \sin \beta + \sin \gamma < 3 \sin \gamma \quad \dots(1)$$

$$\text{Also } \cos \alpha > \cos \beta \geq \cos \gamma \Rightarrow 3 \cos \alpha > \cos \alpha + \cos \beta + \cos \gamma > 3 \sin \gamma$$

$$\frac{1}{3 \cos \alpha} < \frac{1}{\cos \alpha + \cos \beta + \cos \gamma} < \frac{1}{3 \cos \gamma} \quad \dots(2)$$

Multiplying (1) and (2), we get

$$\tan \alpha < \frac{\sin \alpha + \sin \beta + \sin \gamma}{\cos \alpha + \cos \beta + \cos \gamma} < \tan \gamma$$

[Tangents and Normals]

25. If a variable tangent to the curve $x^2y = c^3$ makes intercepts a, b on x and y axis respectively, then the value of a^2b is

- (1) $27c^3$ (2) $\frac{4}{27}c^3$ (3) $\frac{27}{4}c^3$ (4) $\frac{4}{9}c^3$

Sol. Answer (3)

$$x^2y = c^3$$

$$\Rightarrow y = \frac{c^3}{x^2}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{2c^3}{x^3}$$

Variable tangent :

$$(y - y_1) = -\frac{2c^3}{x_1^3}(x - x_1)$$

$$a = \frac{y_1 x_1^3}{2c^3} + x_1$$

$$= \frac{x_1}{2} + x_1$$

$$= \frac{3x_1}{2}$$

$$b = y_1 + \frac{2c^3}{x_1^2} = \frac{3c^3}{x_1^2}$$

$$\Rightarrow a^2b = \frac{9}{4}x_1^2 \times \frac{3c^3}{x_1^2} = \frac{27}{4}c^3$$

26. Let C be the curve $y = x^3$ (where x takes all real values). The tangent at A except (0, 0) meets the curve again at B. If the gradient at B is k times the gradient at A, then k is equal to

(1) 4

(2) 2

(3) -2

(4) $\frac{1}{4}$

Sol. Answer (1)

Tangent at A

$$(y - y_1) = 3x_1^2(x - x_1)$$

$$y - x_1^3 = 3x_1^2x - 3x_1^3$$

$$y = 3x_1^2x - 2x_1^3$$

Solving tangent with curve

$$3x_1^2x - 2x_1^3 = x^3$$

$$\text{Or, } x^3 - 3x_1^2x + 2x_1^3 = 0$$

$$\Rightarrow (x - x_1)(x_2 + x x_1 - 2x_1^2) = 0$$

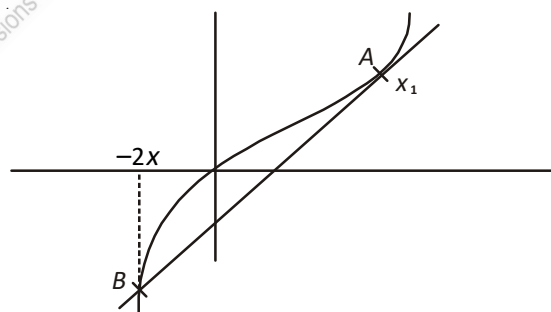
$$\Rightarrow (x - x_1)^2(x + 2x_1) = 0$$

$$\Rightarrow x = (x_1, -2x_1)$$

$$\therefore \text{Gradient at A} = 3x_1^2$$

$$\text{Gradient at B} = 3(4x_1^2)$$

$$\therefore k = 4$$



27. A curve is represented by the equations,

$$x = \sec^2 t \text{ and } y = \cot t$$

where t is a parameter. If the tangent at the point P on the curve where $t = \frac{\pi}{4}$ meets the curve again at the point Q then, $|PQ|$ is equal to :

(1) $\frac{5\sqrt{3}}{2}$

(2) $\frac{5\sqrt{5}}{2}$

(3) $\frac{2\sqrt{5}}{3}$

(4) $\frac{3\sqrt{5}}{2}$

Sol. Answer (4)

$$x = \sec^2 t \Rightarrow \frac{dx}{dt} = 2 \sec t \cdot \sec t \cdot \tan t$$

$$y = \cot t \Rightarrow \frac{dy}{dt} = -\operatorname{cosec}^2 t$$

$$\frac{dy}{dx} = \frac{-\operatorname{cosec}^2 t}{2 \frac{\sin t}{\cos^3 t}}$$

$$\Rightarrow \left(\frac{dy}{dx} \right) = -\frac{1}{2} (\cot t)^3$$

$$\Rightarrow \left(\frac{dy}{dx} \right)_{t=\frac{\pi}{4}} = -\frac{1}{2}$$

Equation of tangent

$$y - 1 = -\frac{1}{2}(x - 2)$$

$$\Rightarrow 2y - 2 = -x + 2$$

$$\Rightarrow x + 2y = 4$$

Equation of curve

$$x - \frac{1}{y^2} = 1$$

$$\Rightarrow -\frac{1}{y^2} = 1 - x$$

$$\Rightarrow y^2 = -\frac{1}{1-x}$$

$$\Rightarrow (2y - 3)y^2 = -1$$

$$\Rightarrow 2y^3 - 3y^2 + 1 = 0$$

For point Q,

$$\Rightarrow y = \frac{-1}{2}, x = 5$$

$$\text{Distance } |PQ| = \sqrt{(5-2)^2 + \left(-\frac{1}{2}-1\right)^2} = \sqrt{9 + \frac{9}{4}} = \sqrt{\frac{45}{4}} = \frac{3\sqrt{5}}{2}$$

28. If $x = 2t - 3t^2$ and $y = 6t^3$ then $\frac{dy}{dx}$ at point $(-1, 6)$ is

(1) $\frac{1}{3}$

(2) $-\frac{9}{2}$

(3) $\frac{8}{3}$

(4) 0

Sol. Answer (2)

$$\frac{dx}{dt} = 2 - 6t$$

$$\frac{dy}{dt} = 18t^2$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{18t^2}{2-6t}$$

$$\text{If } y = 6 = 6t^3$$

$$\Rightarrow t = 1$$

$$\left(\frac{dy}{dx}\right)_{t=1} = \frac{18 \times 1^2}{2-6 \times 1} = \frac{18}{-4} = -\frac{9}{2}$$

29. Find the equation of tangent to the curve $y = \sqrt{x}$ at the point where tangent drawn makes an angle $\frac{\pi}{4}$ with '+' x-axis :

(1) $x - y = -\frac{1}{4}$

(2) $x - y = \frac{1}{4}$

(3) $x + y = \frac{1}{2}$

(4) $x - y = \frac{1}{2}$

Sol. Answer (1)

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$

$$\frac{1}{2\sqrt{x}} = 1$$

$$\Rightarrow \sqrt{x} = \frac{1}{2}$$

$$\Rightarrow x = \frac{1}{4}$$

$$y = \frac{1}{2}$$

$$\Rightarrow \left(x - \frac{1}{4}\right) = y - \frac{1}{2}$$

$$\Rightarrow x - y = -\frac{1}{4}$$

30. The co-ordinates of the point on the curve $9y^2 = x^3$ where the normal of positive slope to the curve makes equal intercepts with the axis is

(1) $\left(1, \frac{1}{3}\right)$

(2) $(3, \sqrt{3})$

(3) $\left(4, \frac{8}{3}\right)$

(4) $\left(\frac{6}{5}, \frac{2}{5}\sqrt{\frac{6}{5}}\right)$

Sol. Answer (3)

$$9y^2 = x^3$$

$$\Rightarrow 18y \frac{dy}{dx} = 3x^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2}{6y} = 1$$

$$\Rightarrow \frac{x^3}{6xy} = 1$$

$$\Rightarrow \frac{9y^2}{6xy} = 1$$

$$\Rightarrow 3y = 2x$$

$$\Rightarrow 9y^2 = \left(\frac{3}{2}y\right)^3$$

$$\Rightarrow 9y^2 = \frac{27}{8}y^3$$

$$\Rightarrow 8y^2 - 3y^3 = 0$$

$$\Rightarrow y^2(8 - 3y) = 0$$

$$\Rightarrow y = \frac{8}{3}$$

31. The tangent of angle made by the tangent of the curve $x = a(t + \sin t \cos t)$, $y = a(1 + \sin t)^2$ with the x-axis at any point on it is

(1) $\frac{1}{4}(\pi + 2t)$

(2) $\frac{1 + \sin t}{\cos t}$

(3) $\frac{1}{4}(2t - \pi)$

(4) $\frac{1 + \sin t}{\cos 2t}$

Sol. Answer (2)

$$x = a(t + \sin t \cos t)$$

$$y = a(1 + \sin t)^2$$

$$\frac{dx}{dt} = a(1 + \cos 2t)$$

$$\frac{dy}{dt} = 2a(1 + \sin t) \cos t$$

$$\frac{dy}{dx} = \frac{2a(1 + \sin t) \cdot \cos t}{a(1 + \cos 2t)} = \frac{2(1 + \sin t) \cdot \cos t}{2 \cos^2 t}$$

$$\tan \theta = \left(\frac{1 + \sin t}{\cos t} \right)$$

32. Consider the curve represented parametrically by the equation

$$x = t^3 - 4t^2 - 3t \text{ and}$$

$$y = 2t^2 + 3t - 5 \text{ where } t \in R$$

If H denotes the number of point on the curve where the tangent is horizontal and V the number of point where the tangent is vertical then

(1) $H = 2$ and $V = 1$

(2) $H = 1$ and $V = 2$

(3) $H = 2$ and $V = 2$

(4) $H = 1$ and $V = 1$

Sol. Answer (2)

$$x = t^3 - 4t^2 - 3t$$

$$y = 2t^2 + 3t - 5$$

$$\begin{aligned} \Rightarrow \frac{dx}{dt} &= 3t^2 - 8t - 3 \\ &= 3t^2 - 9t + t - 3 \\ &= (3t + 1)(t - 3) \end{aligned}$$

$$\Rightarrow \frac{dy}{dt} = 4t + 3$$

$$\frac{dy}{dx} = \frac{(4t + 3)}{(3t + 1)(t - 3)}$$

For vertical tangents $\frac{dy}{dx} \rightarrow \infty \Rightarrow t = 3, -\frac{1}{3}$

For Horizontal tangents $\frac{dy}{dx} \rightarrow 0 \Rightarrow t = -\frac{3}{4}$

So, $H = 1, V = 2$

33. The point on the curve $y = 6x - x^2$ where the tangent is parallel to x-axis is

(1) (0, 0)

(2) (2, 8)

(3) (6, 0)

(4) (3, 9)

Sol. Answer (4)

$$y = 6x - x^2$$

$$\Rightarrow \frac{dy}{dx} = 6 - 2x = 0$$

$$x = 3$$

$$y = 18 - 9 = 9$$

$$\text{i.e., } (3, 9)$$

34. For the curve

$$y = 3\sin\theta \cdot \cos\theta, \quad x = e^\theta \sin \theta, \quad 0 \leq \theta \leq \pi,$$

the tangent is parallel to x-axis when θ is

- (1) 0 (2) $\frac{\pi}{2}$ (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{6}$

Sol. Answer (3)

$$y = 3 \sin \theta \cdot \cos \theta$$

$$= \frac{3}{2} \sin 2\theta$$

$$\Rightarrow \frac{dy}{d\theta} = 3 \cos 2\theta$$

$$x = e^\theta \sin \theta$$

$$\Rightarrow \frac{dx}{d\theta} = e^\theta \cdot \cos \theta + e^\theta \sin \theta$$

$$\frac{dx}{d\theta} = e^\theta (\sin \theta + \cos \theta)$$

$$\frac{dy}{dx} = \frac{3 \cos 2\theta}{e^\theta (\sin \theta + \cos \theta)} = 0, \text{ when } \theta = \frac{\pi}{4}$$

35. The slope of normal to the curve $x^3 = 8a^2y$, $a > 0$ at a point in the first quadrant is $-\frac{2}{3}$, then point is

- (1) $(2a, -a)$ (2) $(2a, a)$ (3) $(a, 2a)$ (4) (a, a)

Sol. Answer (2)

$$x^3 = 8a^2y$$

$$\Rightarrow 3x^2 = 8a^2 \frac{dy}{dx}, \quad a > 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{3x^2}{8a^2} = \frac{3}{2}$$

$$\Rightarrow x^2 = 4a^2 \quad (\text{when } x = 2a, y = a)$$

$$\Rightarrow x = \pm 2a$$

36. The normal to the curve $x = a(\cos\theta + \theta\sin\theta)$,
 $y = a(\sin\theta - \theta\cos\theta)$ at any point ' θ ' such that

(1) It passes through $\left(\frac{a\pi}{2}, -a\right)$

(2) Is at a constant distance from origin

(3) It passes through origin

(4) It makes an angle $\left(\frac{\pi}{2} - \theta\right)$ with the '+' x-axis

Sol. Answer (2)

$$x = a(\cos\theta + \theta\sin\theta)$$

$$\Rightarrow \frac{dx}{d\theta} = a(-\sin\theta + \theta\cos\theta + \sin\theta)$$

$$= a\theta\cos\theta$$

$$y = a(\sin\theta - \theta\cos\theta)$$

$$\Rightarrow \frac{dy}{d\theta} = a(\cos\theta + \theta\sin\theta - \cos\theta) = a\theta\sin\theta$$

$$\frac{dy}{dx} = \tan\theta$$

$$\Rightarrow [y - a(\sin\theta - \theta\cos\theta)] = \frac{-\cos\theta}{\sin\theta} [x - a(\cos\theta + \theta\sin\theta)]$$

$$\Rightarrow y\sin\theta + x\cos\theta = a[(\cos\theta + \theta\sin\theta) \cdot \cos\theta + \sin\theta(\sin\theta - \theta\cos\theta)]$$

$$= a[1]$$

$$x\cos\theta + y\sin\theta = a$$

37. The equation of tangent to the curve $x = a\cos^3 t$, $y = a\sin^3 t$ at ' t ' is

(1) $x\sec t - y\operatorname{cosec} t = a$

(2) $x\sec t + y\operatorname{cosec} t = a$

(3) $x\operatorname{cosec} t + y\cos t = a$

(4) $x\sec t + y\cos t = -a$

Sol. Answer (2)

$$x = a\cos^3 t$$

$$y = a\sin^3 t$$

$$\frac{dx}{dt} = -3a\cos^2 t \cdot \sin t$$

$$\frac{dy}{dt} = 3a\sin^2 t \cdot \cos t$$

$$\frac{dy}{dx} = -\frac{\sin^2 t \cdot \cos t}{\cos^2 t \cdot \sin t} = -\tan t$$

Equation of tangent

$$(y - a \sin^3 t) = -\tan t (x - a \cos^3 t)$$

$$\Rightarrow y \cos t - a \sin^3 t \cdot \cos t = -x \sin t + a \cos^3 t \cdot \sin t$$

$$\Rightarrow x \sin t + y \cos t = a \sin t \cdot \cos t$$

$$\Rightarrow x \sec t + y \operatorname{cosec} t = a$$

38. The slope of the tangent to the curve $y = \int_0^x \frac{dx}{1+x^3}$ at the point where $x = 1$ is

(1) $\frac{1}{2}$

(2) 1

(3) $\frac{1}{4}$

(4) $\frac{1}{5}$

Sol. Answer (1)

The equation of the curve is

$$y = \int_0^x \frac{dx}{1+x^3}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1+x^3} \Rightarrow \left(\frac{dy}{dx} \right)_{x=1} = \frac{1}{1+1} = \frac{1}{2}$$

Slope of the tangent to the curve at $x = 1$ is $\frac{1}{2}$

39. If at each point of the curve $y = x^3 - ax^2 + x + 1$ the tangent is inclined at an acute angle with the positive direction of the x -axis then

(1) $a > 0$

(2) $a \leq \sqrt{3}$

(3) $(-\sqrt{3} < a < \sqrt{3})$

(4) $2 \leq a \leq 3$

Sol. Answer (3)

We have, $y = x^3 - ax^2 + x + 1$

$$\Rightarrow \frac{dy}{dx} = 3x^2 - 2ax + 1$$

tangent make an acute angle with positive direction of x -axis

$$\text{Hence } \frac{dy}{dx} \geq 0$$

$$\Rightarrow 3x^2 - 2ax + 1 \geq 0, \forall x \in R \Rightarrow 4a^2 - 12 \leq 0$$

$$\Rightarrow -\sqrt{3} \leq a \leq \sqrt{3} \Rightarrow a \in [-\sqrt{3}, \sqrt{3}]$$

40. If m be the slope of a tangent to the curve $e^y = 1 + x^2$ then

(1) $|m| > 1$

(2) $m < 1$

(3) $|m| < 1$

(4) $|m| \leq 1$

Sol. Answer (4)

The equation of the given curve is

$$e^y = 1 + x^2$$

$$\Rightarrow e^y \frac{dy}{dx} = 2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x}{1+x^2}$$

$$\Rightarrow m = \frac{2x}{1+x^2} = \text{Slope of a tangent to the given curve}$$

$$\text{Since } -1 \leq \frac{2x}{1+x^2} \leq 1$$

$$\text{Hence } |m| \leq 1.$$

41. The normal to the curve $2x^2 + y^2 = 12$ at the point $(2, 2)$ cuts the curve again at

(1) $\left(-\frac{22}{9}, -\frac{2}{9}\right)$

(2) $\left(\frac{22}{9}, \frac{2}{9}\right)$

(3) $(-2, -2)$

(4) $(0, \sqrt{12})$

Sol. Answer (1)

$$\frac{x^2}{6} + \frac{y^2}{12} = 1 \quad \dots(1)$$

The equation of the normal to the given ellipse at $(2, 2)$ is given by

$$\left(\frac{x-x_1}{x_1}\right)a^2 = \left(\frac{y-y_1}{y_1}\right)b^2$$

$$\Rightarrow \frac{x-2}{2} \times 6 = \frac{y-2}{2} \times 12$$

$$\Rightarrow x - 2y + 2 = 0 \quad \dots(2)$$

On solving (1) and (2) we get

$$\text{The point of intersection as } \left(-\frac{22}{9}, -\frac{2}{9}\right).$$

42. The tangent to the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at any point on it cuts the axes Ox and Oy at P and Q respectively then $OP + OQ$ is

(1) $2a$

(2) a

(3) $\frac{a}{2}$

(4) $2\sqrt{a}$

Sol. Answer (2)Let point be (x_1, y_1)

$$\sqrt{x} + \sqrt{y} = \sqrt{a} \Rightarrow \sqrt{x_1} + \sqrt{y_1} = \sqrt{a} \Rightarrow \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = -\sqrt{\frac{y_1}{x_1}}$$

$$\text{The equation of tangent } y - y_1 = -\sqrt{\frac{y_1}{x_1}}(x - x_1)$$

$$\frac{x}{\sqrt{x_1}} + \frac{y}{\sqrt{y_1}} = \sqrt{x_1} + \sqrt{y_1}$$

$$OP = \sqrt{x_1}(\sqrt{x_1} + \sqrt{y_1})$$

$$OQ = \sqrt{y_1}(\sqrt{x_1} + \sqrt{y_1})$$

$$OP + OQ = (\sqrt{x_1} + \sqrt{y_1})^2 = a$$

43. The slope of tangent to the curve represented by $x = t^2 + 3t - 8$ and $y = 2t^2 - 2t - 5$ at the point $M(2, -1)$ is

(1) $\frac{7}{6}$

(2) $\frac{2}{3}$

(3) $\frac{3}{2}$

(4) $\frac{6}{7}$

Sol. Answer (4)

$$\text{We have, } x = t^2 + 3t - 8, \quad y = 2t^2 - 2t - 5$$

at $M(2, -1)$

$$y = 2t^2 - 2t - 5 = -1$$

$$\Rightarrow t^2 - t - 2 = 0 \Rightarrow (t - 2)(t + 1) = 0 \Rightarrow t = 2, t = -1$$

$$\text{Also } x = t^2 + 3t - 8 = 2 \Rightarrow t^2 + 3t - 10 = 0$$

$$\Rightarrow (t + 5)(t - 2) = 0 \Rightarrow t = 2, -5$$

The common value of t is 2

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{4t - 2}{2t + 3}$$

$$m = \left(\frac{dy}{dx}\right)_{t=2} = \frac{6}{7}$$

44. The point(s) on the curve $y^3 + 3x^2 = 12y$, where the tangent is vertical is (are)

(1) $(\pm 4\sqrt{3}, -2)$

(2) $\left(\pm\sqrt{\frac{11}{3}}, 1\right)$

(3) $(0, 0)$

(4) $\left(\pm\frac{4}{\sqrt{3}}, 2\right)$

Sol. Answer (4)

We have, $y^3 + 3x^2 = 12y$... (1)

$$\Rightarrow \frac{dy}{dx} = -\frac{2x}{y^2 - 4}$$

Let point be (x_1, y_1)

$$m = -\frac{2x_1}{y_1^2 - 4}$$

Tangent is vertical hence $\frac{dx}{dy} = 0$

$$\Rightarrow y_1 = \pm 2$$

Let us put $y = \pm 2$ in (1)

$$\text{If } y_1 = 2 \Rightarrow x_1 = \pm \frac{4}{\sqrt{3}}$$

Hence points on the given curve where tangents are vertical are $\left(\pm \frac{4}{\sqrt{3}}, 2\right)$.

45. The number of tangents to the curve $x^{3/2} + y^{3/2} = a^{3/2}$, where the tangents are equally inclined to axes, is

(1) 2

(2) 1

(3) 0

(4) 4

Sol. Answer (2)

The equation of the curve is

$$x^{3/2} + y^{3/2} = a^{3/2} \quad \dots (1)$$

$$\Rightarrow \frac{dy}{dx} = -\sqrt{\frac{x}{y}}$$

Since the tangent is equally inclined to the axes

$$\text{hence } \frac{dy}{dx} = \pm 1$$

$$\Rightarrow \sqrt{\frac{x}{y}} = \pm 1 \quad \Rightarrow \frac{x}{y} = 1 \quad \Rightarrow y = x$$

Putting $x = y$ in (1), we get,

$$2x^{3/2} = a^{3/2}$$

$$\Rightarrow x = \left(\frac{1}{2}\right)^{2/3} a \quad \Rightarrow y = \left(\frac{1}{2}\right)^{2/3} a$$

Hence number of tangent = 1

46. The number of tangents to the curve

$$y^2 - 2x^3 - 4y + 8 = 0 \text{ that pass through } (1, 2) \text{ is}$$

(1) 3

(2) 1

(3) 2

(4) 6

Sol. Answer (3)

$$y^2 - 2x^3 - 4y + 8 = 0$$

Let point be (x_1, y_1)

$$\left(\frac{dy}{dx}\right)_{(x_1, y_1)} = \frac{3x_1^2}{y_1 - 2}$$

Equation of tangent at (x_1, y_1) is given by

$$y - y_1 = \frac{3x_1^2}{y_1 - 2}(x - x_1)$$

Since the tangent passes through $(1, 2)$

$$\text{Hence, } 2 - y_1 = \frac{3x_1^2}{y_1 - 2}(1 - x_1) \Rightarrow 4 + y_1^2 - 4y_1 = 3x_1^3 - 3x_1^2$$

$$\Rightarrow 3x_1^3 - 3x_1^2 - y_1^2 + 4y_1 - 4 = 0 \quad \dots(1)$$

Also (x_1, y_1) lies on $y^2 - 2x^3 - 4y + 8 = 0$

$$\text{Hence } y_1^2 - 2x_1^3 - 4y_1 + 8 = 0 \quad \dots(2)$$

Solving (1) and (2)

$$x_1^3 - 3x_1^2 + 4 = 0$$

$$(x_1 + 1)(x_1^2 - 4x_1 + 4) = 0$$

$$(x_1 + 1)(x_1 - 2)^2 = 0$$

$$x_1 = -1, 2$$

From (2) at $x = -1$.

$$y_1^2 - 4y_1 + 10 = 0$$

Imaginary roots

at $x = 2$

$$y_1^2 - 4y_1 - 8 = 0$$

$$y_1 = \frac{4 \pm \sqrt{16 + 32}}{2} = 2 \pm \sqrt{3}$$

Points are $(2, 2 \pm \sqrt{3})$ hence number of tangent = 2.

47. The curve $\frac{x^n}{a^n} + \frac{y^n}{b^n} = 2$ touches the line $\frac{x}{a} + \frac{y}{b} = 2$ at the point

- (1) (b, a) (2) (a, b) (3) $(1, 1)$ (4) $\left(\frac{1}{b}, \frac{1}{a}\right)$

Sol. Answer (2)

$$\frac{x^n}{a^n} + \frac{y^n}{b^n} = 2$$

Differentiating it w.r.t. x we get

$$\frac{nx^{n-1}}{a^n} + \frac{ny^{n-1}}{b^n} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{b^n}{a^n} \frac{x^{n-1}}{y^{n-1}}$$

Let the point be (x_1, y_1)

The equation of tangent

$$\frac{xx_1^{n-1}}{a^n} + \frac{yy_1^{n-1}}{b^n} = \frac{x_1^n}{a^n} + \frac{y_1^n}{b^n} \quad \dots(1)$$

According to question tangent is $\frac{x}{a} + \frac{y}{b} = 2 \quad \dots(2)$

Equations (1) & (2) represent same line

$$\text{Hence } \frac{x_1^{n-1}}{a^{n-1}} = \frac{y_1^{n-1}}{b^{n-1}} = \frac{1}{2} \left[\frac{x_1^n}{a^n} + \frac{y_1^n}{b^n} \right]$$

$$\Rightarrow x_1 = a, y_1 = b$$

Hence point is (a, b)

48. Let the equation of a curve be $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$. If θ changes at a constant rate k then the rate of change of the slope of the tangent to the curve at $\theta = \frac{\pi}{3}$ is

- (1) $\frac{2k}{\sqrt{3}}$ (2) $\frac{k}{\sqrt{3}}$ (3) k (4) $\frac{2k}{3}$

Sol. Answer (4)

We have $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$

$$\Rightarrow \frac{dy}{dx} = \tan \frac{\theta}{2} \Rightarrow m = \tan \frac{\theta}{2} = \text{Slope of the tangent to the given curve}$$

$$\Rightarrow \frac{dm}{dt} = \frac{1}{2} \sec^2 \frac{\theta}{2} \frac{d\theta}{dt} \Rightarrow \left(\frac{dm}{dt} \right)_{\theta=\frac{\pi}{3}} = \frac{1}{2} \times \frac{4}{3} k$$

$$\text{The rate of change of the slope of the tangent} = \frac{2k}{3}$$

49. The equation of the common tangent to the curves $y^2 = 8x$ and $xy = -1$ is

(1) $y = x + 2$

(2) $y = 2x + 1$

(3) $2y = x + 8$

(4) $3y = 9x + 2$

Sol. Answer (1)

Equation of tangent to the curve $y^2 = 8x$ is

$$y = mx + \frac{2}{m} \quad \dots(1)$$

It is also the tangent to $xy = -1$

Hence $x\left(mx + \frac{2}{m}\right) = -1$

i.e., $mx^2 + \frac{2}{m}x + 1 = 0$ have equal real roots.

Consequently

$$b^2 - 4ac = 0$$

$$\Rightarrow \frac{4}{m^2} - 4m = 0 \Rightarrow m = 1.$$

Hence equation of the tangent common to the given curves is $y = x + 2$.

50. Let A be the area formed by the positive x -axis and the normal and tangent to the circle $x^2 + y^2 = 4$ at $(1, \sqrt{3})$ then A^2 equals

(1) 9

(2) 12

(3) 15

(4) 18

Sol. Answer (2)

$$x^2 + y^2 = 4$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{y} \Rightarrow \left(\frac{dy}{dx}\right)_{(1, \sqrt{3})} = -\frac{1}{\sqrt{3}}$$

The equation of tangent at $(1, \sqrt{3})$ is

$$y - \sqrt{3} = \frac{-1}{\sqrt{3}}(x - 1)$$

and the point of intersection with x axis is $(4, 0)$

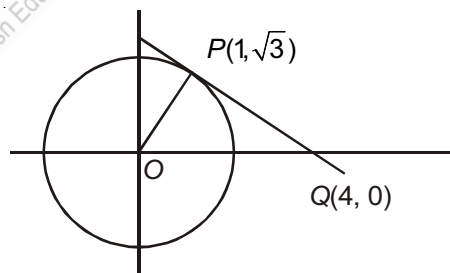
The equation of normal at $(1, \sqrt{3})$ is

$$y - \sqrt{3} = \sqrt{3}(x - 1)$$

and point of intersection with x -axis is $(0, 0)$

$$\text{Hence } A = \frac{1}{2} \times 4 \times \sqrt{3} = 2\sqrt{3}$$

$$A^2 = 12$$



51. If the tangent to the curve $2y^3 = ax^2 + x^3$ at the point (a, a) cuts off intercepts α and β on the co-ordinate axes, (where $\alpha^2 + \beta^2 = 61$) then a^2 equals

(1) 400 (2) 600 (3) 900 (4) 800

Sol. Answer (3)

We have, $2y^3 = ax^2 + x^3$

$$\Rightarrow \frac{dy}{dx} = \frac{2ax + 3x^2}{6y^2} \Rightarrow \left(\frac{dy}{dx} \right)_{(a,a)} = \frac{5}{6}$$

The equation of tangent to the given curve at (a, a) is $y - a = \frac{5}{6}(x - a)$

$$\Rightarrow x \text{ intercept of the tangent} = \alpha = \frac{1}{5}a$$

$$\text{and } y \text{ intercept of the tangent} = \beta = \frac{1}{6}a$$

$$\Rightarrow \alpha^2 + \beta^2 = 61 = \frac{a^2}{25} + \frac{a^2}{36} = \frac{61a^2}{900} \Rightarrow a^2 = 900$$

52. If the curves $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$ and $y^3 = 16x$ intersects at right angles, then $3a^2$ is equal to

(1) 1 (2) 2 (3) 3 (4) 4

Sol. Answer (4)

Let point of intersection be (x_1, y_1)

$$\frac{x^2}{a^2} + \frac{y^2}{4} = 1 \quad \dots(1)$$

$$\Rightarrow \frac{dy}{dx} = -\frac{4}{a^2} \cdot \frac{x}{y}$$

$$m_1 = \left(\frac{dy}{dx} \right)_{(x_1, y_1)} = -\frac{4}{a^2} \cdot \frac{x_1}{y_1}$$

$$y^3 = 16x \quad \dots(2)$$

$$m_2 = \left(\frac{dy}{dx} \right)_{(x_1, y_1)} = \frac{16}{3y_1^2}$$

The two curves intersect at right angles then $m_1 m_2 = -1$

$$\left(\frac{4}{a^2} \frac{x_1}{y_1}\right) \left(\frac{16}{3y_1^2}\right) = 1$$

$$x_1 = \frac{3a^2 y_1^3}{64}$$

$$\text{But } y_1^3 = 16x_1 = 16 \cdot \frac{3a^2 y_1^3}{64}$$

$$y_1^3 = 16x_1 \Rightarrow 3a^2 = 4$$

53. The area of the triangle formed by the tangent to the curve $y = \frac{8}{4+x^2}$ at $x = 2$ and the co-ordinate axes is

(1) 1

(2) 2

(3) 3

(4) 4

Sol. Answer (4)

$$\text{Given curve is } y = \frac{8}{4+x^2}$$

$$\text{When } x = 2, y = 1$$

$$\therefore \frac{dy}{dx} = -\frac{8}{(4+x^2)^2} \times 2x$$

$$\therefore \text{ point is } (2, 1)$$

$$\therefore m = -\frac{32}{64} = -\frac{1}{2}$$

Equation of tangent at $P(2, 1)$ is

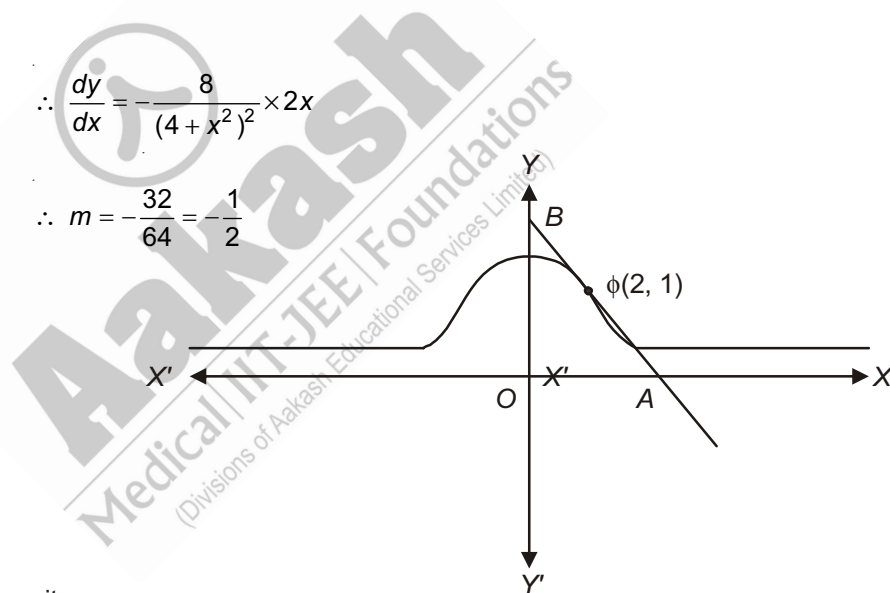
$$y - 1 = -\frac{1}{2}(x - 2)$$

$$\Rightarrow xy - 2 = -x + 2$$

$$\Rightarrow x + xy = 4$$

$$\therefore A = (4, 0), B = (0, 2)$$

$$\therefore \text{ar}(\triangle OAB) = \frac{1}{2} \times 4 \times 2 = 4 \text{ sq. units}$$



[Maxima and Minima]

54. Suppose x_1 and x_2 are the point of maximum and the point of minimum respectively of the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$ respectively, ($a > 0$) then for the equality $x_1^2 = x_2$ to be true the value of 'a' must be

(1) 0

(2) 2

(3) 1

(4) $\frac{1}{4}$

Sol. Answer (2)

$$6(x^2 - 3ax + 2a^2)$$

$$= 6(x - a)(x - 2a); \quad a > 0$$

$x = a$ is point of maxima

$x = 2a$ is point of minima

$$\therefore a^2 = 2a$$

$$\Rightarrow a = 0 \text{ or } a = 2$$

$$\text{but } a > 0 \Rightarrow a = 2$$

55. Point 'A' lies on the curve $y = e^{-x^2}$ and has the coordinate (x, e^{-x^2}) where $x > 0$. Point B has the coordinates $(x, 0)$. If 'O' is the origin, then the maximum area of the triangle AOB is

(1) $\frac{1}{\sqrt{2e}}$

(2) $\frac{1}{\sqrt{4e}}$

(3) $\frac{1}{\sqrt{e}}$

(4) $\frac{1}{\sqrt{8e}}$

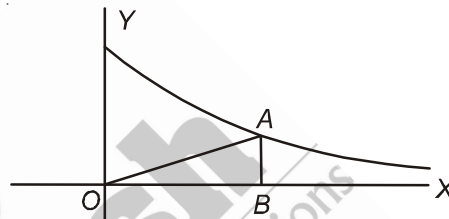
Sol. Answer (4)

$$A = \frac{1}{2} \times x \times e^{-x^2}$$

$$\frac{dA}{dx} = \frac{1}{2} [e^{-x^2} - 2x^2 e^{-x^2}] = 0$$

$$\Rightarrow x^2 = \frac{1}{2}$$

$$\therefore A = \frac{1}{2} \times \frac{1}{\sqrt{2}} \times e^{-\frac{1}{2}} = \frac{1}{\sqrt{8e}}$$



56. The number of points of maxima/minima of $f(x) = x(x+1)(x+2)(x+3)$ is

(1) 0

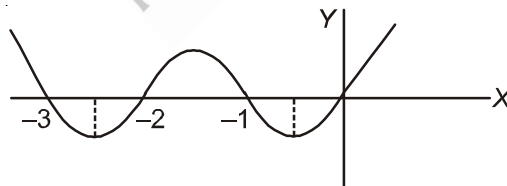
(2) 1

(3) 2

(4) 3

Sol. Answer (4)

$$y = x(x+1)(x+2)(x+3)$$



Clearly, option (4).

57. The difference between the greatest and the least values of the function,

$$f(x) = \sin 2x - x \text{ on } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \text{ is}$$

(1) π

(2) $\left(\sqrt{3} - \frac{\pi}{3}\right)$

(3) $\frac{\sqrt{3}}{2} + \frac{\pi}{3}$

(4) $-\frac{\sqrt{3}}{2} + \frac{2\pi}{3}$

Sol. Answer (1)

$$y = \sin 2x - x$$

$$y' = 2 \cos 2x - 1 = 0$$

$$\Rightarrow \cos 2x = \frac{1}{2}$$

$$\Rightarrow 2x = \pm \frac{\pi}{3}$$

$$\Rightarrow x = \pm \frac{\pi}{6}$$

$$y\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

$$y\left(-\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2} + \frac{\pi}{6}$$

$$-\frac{\sqrt{3}}{2} + \frac{\pi}{6} = -\frac{\pi}{2}$$

$$y\left(-\frac{\pi}{2}\right) = \frac{\pi}{2}$$

$$\Rightarrow y_{\max} = \frac{\pi}{2}$$

$$y_{\min} = -\frac{\pi}{2}$$

$$\therefore \text{Difference} = \pi$$

58. Difference between the greatest and the least values of the function

$$f(x) = x (\ln x - 2) \text{ on } [1, e^2] \text{ is}$$

(1) 2

(2) e

(3) e^2

(4) 1

Sol. Answer (2)

$$f(x) = x (\ln x - 2)$$

$$f'(x) = (\ln x - 2) + 1$$

$$= \ln x - 1$$

$$= 0$$

$$\Rightarrow x = e$$

$$f''(x) = \frac{1}{x}$$

$$\Rightarrow f''(e) = \frac{1}{e} > 0$$

$\therefore x = e$ is minima

$$\therefore \text{Minimum value} = e(\ln e - 2) = -e$$

$$f(1) = 1(0 - 2) = -2$$

$$f(e^2) = e^2(2 - 2) = 0$$

$$\Rightarrow \text{Maximum} = 0$$

$$\therefore \text{Difference} = 0 - (-e) = e$$

59. The sum of lengths of the hypotenuse and another side of a right angle triangle is given. The area of the triangle will be maximum if the angle between them is

(1) $\frac{\pi}{6}$

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{3}$

(4) $\frac{5\pi}{12}$

Sol. Answer (3)

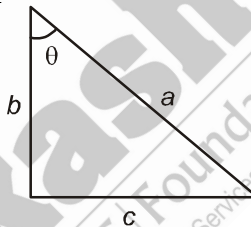
$$A = \frac{1}{2}bc$$

$$A = \frac{1}{2}b\sqrt{a^2 - b^2}$$

$$A = \frac{1}{2}b\sqrt{(b+k)^2 - b^2} \text{ where } a + b = k$$

$$\frac{dA}{db} = 0 \Rightarrow b = \frac{k}{3} \text{ and } a = \frac{2k}{3}$$

$$\text{Hence, } \cos \theta = \frac{1}{2}, \theta = \frac{\pi}{3}$$



60. Divide 64 into two parts such that sum of the cubes of two parts is minimum. The two parts are

(1) 44, 20

(2) 16, 48

(3) 32, 32

(4) 50, 14

Sol. Answer (3)

$$x + y = 64$$

$$S = x^3 + y^3$$

$$S = x^3 + (64 - x)^3$$

$$S' = 3x^2 - 3(64 - x)^2$$

$$\text{For maxima and minima, } S' = 0$$

$$\Rightarrow x^2 = (64 - x)^2$$

$$\Rightarrow x = 64 - x$$

$$\Rightarrow x = 32$$

61. If $f(x) = x^5 - 5x^4 + 5x^3 - 10$ has local maximum and minimum at $x = a$ and $x = b$ respectively then (a, b) is
- (1) $(0, 1)$ (2) $(1, 3)$ (3) $(1, 0)$ (4) $(2, 4)$

Sol. Answer (2)

$$f(x) = x^5 - 5x^4 + 5x^3 - 10$$

$$f'(x) = 5x^4 - 20x^3 + 15x^2$$

For maxima and minima, $f'(x) = 0$

$$5x^2 (x^2 - 4x + 3) = 0$$

$$x^2 (x - 3) (x - 1) = 0$$

$x = 0$ neither maxima nor minima

$$x = 3, 1$$

$$f''(x) = 20x^3 - 60x^2 + 30x$$

$$= 10x (2x^2 - 6x + 3)$$

$f''(3) > 0$ Point of minima

$f''(1) < 0$ Point of maxima

62. The real number x when added to its reciprocal gives the minimum value of sum at x equal to ($x > 0$)
- (1) 1 (2) -1 (3) -2 (4) 2

Sol. Answer (1)

$$y = x + \frac{1}{x}, x > 0 \left(x + \frac{1}{x} \geq 2 \text{ if } x > 0 \text{ and minima occurs at } x = 1 \right)$$

63. For all $a, b \in \mathbb{R}$, the function $f(x) = 3x^4 - 4x^3 + 6x^2 + ax + b$ has
- (1) No extremum (2) Exactly one extremum
- (3) Exactly two extremum (4) Three extremum

Sol. Answer (2)

$$f'(x) = 12x^3 - 12x^2 + 12x + a$$

$$= 12x (x^2 - x + 1) + a$$

$$f''(x) = 12 (3x^2 - 2x + 1)$$

$$f''(x) > 0$$

$f'(x)$ is always increasing

So, there is one point at which $f'(x) = 0$.

64. Maximum area of a rectangle which can be inscribed in a circle of a given radius R is

(1) $\frac{\pi R^2}{3}$ (2) $\frac{3R^2}{\sqrt{2}}$ (3) $2R^2$ (4) $3R^2$

Sol. Answer (3)

Let rectangle has width b and height h

$$A = bh$$

$$\text{also, } b^2 + h^2 = (2R)^2 = 4R^2$$

$$\Rightarrow b = \sqrt{4R^2 - h^2}$$

$$\text{so, area } A(h) = h\sqrt{4R^2 - h^2}$$

Area is maximum when A^2 is maxima.

$$A^2 = h^2(4R^2 - h^2)$$

$$f(h) = h^2(4R^2 - h^2)$$

$$\text{For maxima } f'(h) = 0$$

$$\Rightarrow f'(h) = h^2(-2h) + (4R^2 - h^2)2h = 0$$

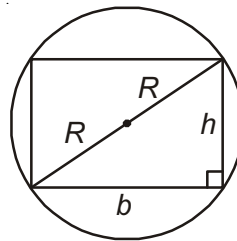
$$\Rightarrow -h^2 + 4R^2 - h^2 = 0$$

$$\Rightarrow h^2 = 2R^2$$

$$\Rightarrow h = \sqrt{2}R$$

From physical nature of problem it is clear that this should be maximum area since minimum area will tend towards zero.

$$\begin{aligned} A_{\max} &= \sqrt{2}R \sqrt{4R^2 - 2R^2} \\ &= \sqrt{2}R \cdot \sqrt{2}R = 2R^2 \end{aligned}$$



65. If atmosphere pressure at a height of h units is given by the function $P(h) = he^{-h}$ then pressure is maximum at the height of

- (1) 1 unit (2) $\frac{1}{\sqrt{e}}$ units (3) e units (4) 2 units

Sol. Answer (1)

$$P(h) = he^{-h}$$

$$\Rightarrow P'(h) = -he^{-h} + e^{-h} = e^{-h}(1 - h)$$

$$P'(h) = 0 \Rightarrow h = 1$$

at $h = 1$, $P'(h)$ changes sign from positive to negative so it is point of maxima.

66. The radius of cylinder of maximum volume which can be inscribed in a right circular cone of radius R and height H (axis of cylinder and cone are same) H given by

- (1) $\frac{R}{2}$ (2) $\frac{R}{3}$ (3) $\frac{2R}{3}$ (4) $\frac{2R}{5}$

Sol. Answer (3)

Let cylinder has radius r and height h .

$\triangle AO'B$ and $\triangle AOC$ are similar

$$\text{so, } \frac{AO'}{AO} = \frac{O'B}{OC}$$

$$\Rightarrow \frac{H-h}{H} = \frac{r}{R}$$

$$\Rightarrow 1 - \frac{h}{H} = \frac{r}{R}$$

$$\Rightarrow \frac{h}{H} = 1 - \frac{r}{R}$$

$$\Rightarrow h = H \left(1 - \frac{r}{R} \right)$$

$$V = \pi r^2 h$$

$$V(r) = \pi r^2 \cdot H \left(1 - \frac{r}{R} \right)$$

$$V(r) = \pi H \left(r^2 - \frac{r^3}{R} \right)$$

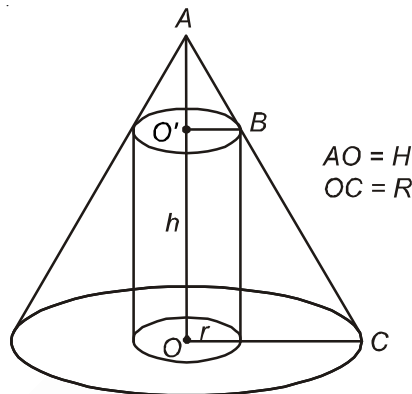
For maximum volume

$$V'(r) = \pi H \left(2r - \frac{3r^2}{R} \right) = 0$$

$$\Rightarrow 2r - \frac{3r^2}{R} = 0$$

$$\Rightarrow 2 - \frac{3r}{R} = 0 \quad (r \neq 0)$$

$$\Rightarrow r = \frac{2R}{3}$$



67. Let $f(x) = \begin{cases} -|x-2| & ; x \leq 3 \\ x^2 - 2x - 4 & ; x > 3 \end{cases}$

Then the number of critical points on the graph of the function is

(1) 1

(2) 2

(3) 3

(4) 4

Sol. Answer (2)

Critical point is the point where $f'(x) = 0$ or $f'(x)$ does not exist.

Function changes its definition at a point $x = 3$ so we will check at $x = 3$

Also, $|x - 2| = (x - 2)$ if $x > 2 = -(x - 2)$ if $x < 2$

$$\text{so, } f(x) = \begin{cases} x - 2 & ; x \leq 2 \\ -(x - 2) & ; 2 < x \leq 3 \\ x^2 - 2x - 4 & ; x > 3 \end{cases}$$

$$f'(x) = \begin{cases} 1 & \text{if } x \leq 2 \\ -1 & \text{if } 2 < x \leq 3 \\ 2x - 2 & \text{if } x > 3 \end{cases}$$

If $f'(x) = 0$ then $x > 3$ and

$$2x - 2 = 0 \Rightarrow x = 1 \notin (3, \infty)$$

so $f'(x)$ cannot be zero anywhere.

Now, $f'(x)$ does not exist at $x = 2$

since, $f'(2 - 0) = 1$

and $f'(2 + 0) = -1$

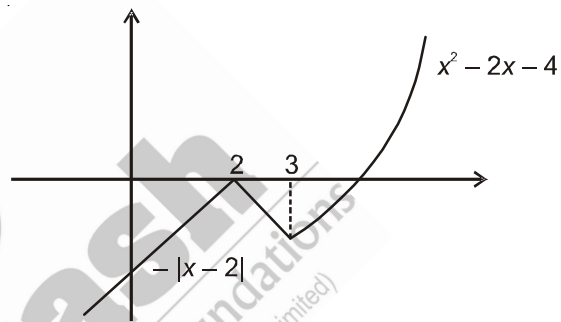
Also, $f'(3 - 0) = -1$

$f'(3 + 0) = 2(3) - 2 = 4$

so, $f'(x)$ does not exist at $x = 3$

Hence, there are two critical points

$x = 2$ and $x = 3$



68. The function ' f ' is defined by $f(x) = x^p (1 - x)^q$ for all $x \in \mathbb{R}$ where p, q are positive integer, has a maximum value, for x equal to

(1) $\frac{pq}{p+q}$

(2) -1

(3) $\frac{1}{2}$

(4) $\frac{p}{p+q}$

Sol. Answer (4)

$$f'(x) = x^p \times q(1 - x)^{q-1}(-1) + p(1 - x)^q x^{p-1}$$

$$= x^p \cdot (1 - x)^q \left[\frac{p}{x} - \frac{q}{(1 - x)} \right]$$

$$f'(x) = 0$$

Then, $x = 1$, (As $x \neq 0$)

$$\text{and } \frac{p}{x} - \frac{q}{1 - x} = 0$$

$$p(1 - x) - qx = 0$$

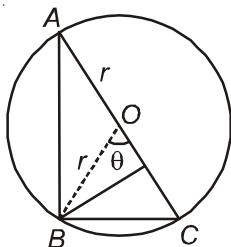
$$p - x(p + q) = 0$$

$$x = \frac{p}{p+q}$$

69. The least area of a circle circumscribing any right triangle of area S is

- (1) πS (2) $2\pi S$ (3) $\sqrt{2}\pi S$ (4) $4\pi S$

Sol. Answer (1)



$$\text{Area of } \triangle ABC = \frac{1}{2} \times 2r \times r \sin \theta$$

$$S = r^2 \sin \theta$$

$$\Rightarrow r^2 = \frac{S}{\sin \theta}$$

$$\text{Area of circle} = \frac{\pi S}{\sin \theta}$$

$$\text{Least area} = \pi S.$$

70. If $f(x) = x(1-x)^3$ then which of following is true?

- (1) $f(x)$ has local maxima at $x = 1$ (2) $f(x)$ has local minima at $x = 1$
 (3) $f(x)$ has local minima at $x = \frac{1}{4}$ (4) $f(x)$ has local maxima at $x = \frac{1}{4}$

Sol. Answer (4)

$$f'(x) = x \cdot 3(1-x)^2(-1) + (1-x)^3 \cdot 1$$

$$\Rightarrow f'(x) = (1-x)^2[-3x+1-x]$$

$$\Rightarrow f'(x) = (1-x)^2[1-4x]$$

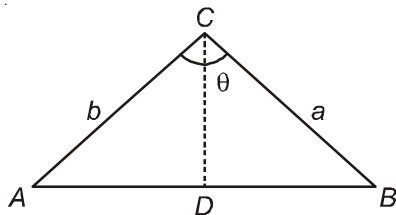
$x = 1$, $f'(x)$ does not change sign so $x = 1$ is not maxima/minima

At $x = \frac{1}{4}$ it has maxima since $f'(x)$ changes sign from positive to negative.

71. Two side of a triangle are to have lengths ' a ' cm and ' b ' cm. If the triangle is to have the maximum area, then the length of the median from the vertex containing the side ' a ' and ' b ' is

- (1) $\frac{1}{2}\sqrt{a^2+b^2}$ (2) $\frac{2a+b}{3}$ (3) $\sqrt{\frac{a^2+b^2}{2}}$ (4) $\frac{a+2b}{3}$

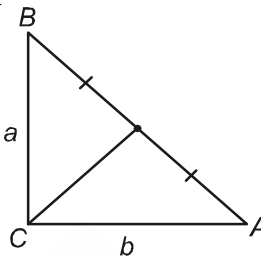
Sol. Answer (1)



$$\text{Area of triangle} = \frac{1}{2}ab \sin \theta$$

For maximum area $\theta = 90^\circ$

$$\text{Length of median } OD = \frac{\sqrt{a^2 + b^2}}{2}$$



72. The cost function at American Gadget is $C(x) = x^3 - 6x^2 + 15x$ (x is thousands of units and $x > 0$). The production level at which average cost is minimum is

(1) 2

(2) 3

(3) 5

(4) 4

Sol. Answer (2)

$$\text{Average cost} = A(x) = \frac{C(x)}{x} = x^2 - 6x + 15$$

$$A'(x) = 2x - 6$$

$$A'(x) = 0$$

$$\Rightarrow x = 3$$

73. Let $f(x) = ax^2 - b|x|$, where a and b are constant. Then at $x = 0$, $f(x)$ has

(1) A maxima whenever $a > 0$, $b > 0$ (2) A maxima whenever $a > 0$, $b < 0$ (3) Minima whenever $a > 0$, $b > 0$ (4) Neither maxima nor minima whenever $a > 0$, $b < 0$

Sol. Answer (1)

$$f(x) = ax^2 - b|x|$$

$$\Rightarrow f'(x) = 2ax - b, x > 0$$

$$= 2ax + b, x < 0$$

74. The maximum value of $\left(\frac{1}{x}\right)^{2x^2}$ is

(1) e (2) $e^{\left(\frac{1}{e}\right)}$

(3) 1

(4) e^e

Sol. Answer (2)

$$y = \left(\frac{1}{x}\right)^{2x^2}$$

$$\Rightarrow y = x^{-2x^2}$$

$$\Rightarrow \frac{dy}{dx} = x^{-2x^2} [-2x - 4x \log x]$$

$$\text{For maxima or minima of } y, \frac{dy}{dx} = 0$$

$$\Rightarrow x = \frac{1}{\sqrt{e}} \text{ as } x \neq 0$$

$$\text{and } \left(\frac{d^2y}{dx^2}\right)_{x=\frac{1}{\sqrt{e}}} < 0$$

$$\text{Hence } y \text{ attains maximum value at } x = \frac{1}{\sqrt{e}} \text{ and } y_{\max} = e^{\frac{1}{e}} = \sqrt[e]{e}$$

75. If $ax + \frac{b}{x} \geq c$, $\forall x > 0$ and a, b, c are positive constants then

$$(1) \quad ab \geq \frac{c^2}{4} \quad (2) \quad ab \leq \frac{c^2}{4} \quad (3) \quad bc \geq \frac{a^2}{4} \quad (4) \quad ac \geq \frac{b^2}{4}$$

Sol. Answer (1)

$$\text{Given } ax + \frac{b}{x} \geq c \quad \forall x > 0$$

A.M. $\geq$ G.M.

$$ax + \frac{b}{x} \geq 2\sqrt{ab} \Rightarrow ab \geq \frac{c^2}{4}$$

Second Method :

$$\text{We have, } ax + \frac{b}{x} \geq c, \forall x > 0 \Rightarrow ax^2 - cx + b \geq 0 \quad \forall x > 0$$

$$\Rightarrow c^2 - 4ab \leq 0 \text{ and } a > 0 \Rightarrow ab \geq \frac{c^2}{4}$$

76. Let x be a number which exceeds its square by the greatest possible quantity. Then x is equal to

$$(1) \quad \frac{1}{2} \quad (2) \quad \frac{1}{4} \quad (3) \quad \frac{3}{4} \quad (4) \quad 1$$

Sol. Answer (1)

$$\text{Let } y = x - x^2 \Rightarrow \frac{dy}{dx} = 1 - 2x$$

$$\text{For maximum or minimum value of } y, \frac{dy}{dx} = 0 \Rightarrow 1 - 2x = 0 \Rightarrow x = \frac{1}{2}$$

$$\text{We observe that } \frac{d^2y}{dx^2} = -2 = \text{negative} \Rightarrow y \text{ is maximum for } x = \frac{1}{2}$$

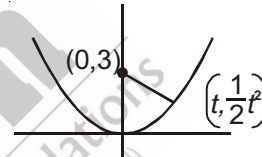
77. The point (0, 3) is nearest to the curve $x^2 = 2y$ at

- (1) $(2\sqrt{2}, 0)$ (2) (0, 0) (3) (2, 2) (4) (2, -2)

Sol. Answer (3)

$$\text{Let point on the curve } x^2 = 2y \text{ be } \left(t, \frac{1}{2}t^2\right)$$

The distance of the point (0, 3) from a point $P\left(t, \frac{1}{2}t^2\right)$ on the given curve $x^2 = 2y$ is



$$d = \sqrt{t^2 + \left(\frac{t^2}{2} - 3\right)^2} \Rightarrow d^2 = t^2 + \left(\frac{t^2}{2} - 3\right)^2$$

Let $d^2 = z$ d is maximum or minimum according as z is maximum or minimum

$$\Rightarrow z = t^2 + \left(\frac{t^2}{2} - 3\right)^2 \Rightarrow \frac{dz}{dt} = 2t + 2t\left(\frac{t^2}{2} - 3\right)$$

$$\text{For maximum or minimum value of } z, \frac{dz}{dt} = 0$$

$$\Rightarrow t = 0, 2, -2 \Rightarrow \frac{d^2z}{dt^2} = 3t^2 - 4$$

$$\frac{d^2z}{dt^2} > 0 \text{ at } t = \pm 2$$

Hence point is (2, 2) or (-2, 2)

Thus the point on the given curve which is nearest from (0, 3) is either (2, 2) or (-2, 2).

78. Let $f(x)$ be a function defined as below :

$$f(x) = \sin(x^2 - 3x), x \leq 0$$

$$= 6x + 5x^2, x > 0$$

then at $x = 0$, $f(x)$

(1) Has a local maximum

(2) Has a local minimum

(3) Is discontinuous

(4) Point of inflexion

Sol. Answer (2)

$$\text{We have, } f(x) = \begin{cases} \sin(x^2 - 3x), & x \leq 0 \\ 6x + 5x^2, & x > 0 \end{cases}$$

We observe that

$$f(0) = 0$$

$$f'(0 + h) = 6 + 10h > 0, \text{ for } h > 0$$

$$\text{and } f'(0 - h) = -(2h + 3) \cos(h^2 + 3h) < 0$$

Here we observe that $f(x)$ changes its sign from -ve to +ve in the neighbourhood of $x = 0$, hence $f(x)$ has a local minimum at $x = 0$

Hence $x = 0$ is a point of local minima.

79. If λ, μ be real numbers such that $x^3 - \lambda x^2 + \mu x - 6 = 0$ has its roots real and positive then minimum value of μ is

(1) $3 \times \sqrt[3]{36}$

(2) 11

(3) 0

(4) 1

Sol. Answer (1)

Let roots of

$$x^3 - \lambda x^2 + \mu x - 6 = 0 \text{ be } \alpha, \beta, \gamma \Rightarrow \alpha + \beta + \gamma = \lambda$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \mu$$

$$\alpha\beta\gamma = 6$$

Applying A.M. – G.M. inequality

$$\frac{\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}}{3} \geq \left(\frac{1}{\alpha} \cdot \frac{1}{\beta} \cdot \frac{1}{\gamma} \right)^{1/3}$$

$$\Rightarrow \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{3} \geq (\alpha\beta\gamma)^{2/3}$$

$$\Rightarrow \mu \geq 3(6)^{2/3}$$

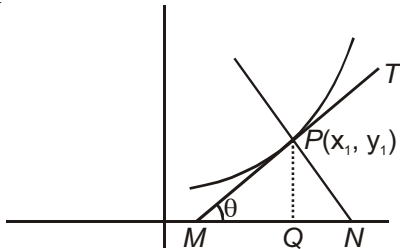
$$\Rightarrow \mu \geq 3(36)^{1/3}$$

$$\text{Minimum } \mu = 3(36)^{1/3}$$

80. If the least area of triangle formed by tangent, normal at any point P on the curve $y = f(x)$ and x -axis is 4 sq. unit. Then the ordinate of the point P (P lies in first quadrant) is

- (1) 1 (2) $\frac{1}{2}$ (3) $\frac{1}{4}$ (4) 2

Sol. Answer (4)



From figure $PQ = y_1$

$$MQ = y_1 \cot \theta$$

$$QN = y_1 \tan \theta \quad \therefore MN = MQ + QN = y_1(\cot \theta + \tan \theta)$$

$$\text{Area of } \triangle PMN = \frac{1}{2} MN \times PQ = \frac{1}{2} y_1^2 (\tan \theta + \cot \theta) \geq \frac{1}{2} y_1^2 \cdot 2 = y_1^2$$

$$\text{Thus } 4 = y_1^2 \quad \Rightarrow y_1 = 2$$

81. If $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$ are such that, $\min f(x) > \max g(x)$, then

- (1) $0 < c < \frac{b}{2}$ (2) $|c| < |b| \sqrt{2}$ (3) $|c| > |b| \sqrt{2}$ (4) $|b| < |c| \sqrt{2}$

Sol. Answer (3)

We have, $f(x) = x^2 + 2bx + 2c^2$, and $g(x) = -x^2 - 2cx + b^2$

$$f(x) = (x+b)^2 + 2c^2 - b^2 \text{ and } g(x) = b^2 + c^2 - (x-c)^2$$

$$\Rightarrow \min f(x) = 2c^2 - b^2 \text{ and } \max g(x) = c^2 + b^2$$

$$\text{Since } \min f(x) > \max g(x)$$

$$\text{Hence } 2c^2 - b^2 > c^2 + b^2$$

$$\Rightarrow c^2 > 2b^2$$

$$\Rightarrow |c| > |b| \sqrt{2}$$

82. If $x \in [-1, 1]$ then the minimum value of

$$f(x) = x^2 + x + 1 \text{ is}$$

- (1) $\frac{3}{4}$ (2) 1 (3) 3 (4) $-\frac{3}{4}$

Sol. Answer (1)

$$f(x) = x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4}, \forall x \in R$$

$$\text{Minimum } f(x) = \frac{3}{4}$$

83. The maximum value of $\cos[\cos(\sin x)]$ is

(1) $\cos(\cos 1)$

(2) $\cos 1$

(3) 1

(4) 0

Sol. Answer (1)

$$f(x) = \cos[\cos(\sin x)]$$

$$f'(x) = \sin[\cos(\sin x)] \sin(\sin x) \cos x$$

for maxima–minima of $f(x)$

$$f'(x) = 0 \Rightarrow x = (2n+1)\frac{\pi}{2}, m\pi$$

$$\Rightarrow f''(x) < 0 \text{ at } x = (2n+1)\frac{\pi}{2} \text{ i.e., } f(x) \text{ is maximum at } x = (2n+1)\frac{\pi}{2}$$

$$\Rightarrow \text{maximum value of } f(x) = \cos(\cos 1)$$

84. The minimum value of $64 \sec \theta + 27 \operatorname{cosec} \theta$ where $\theta \in \left(0, \frac{\pi}{2}\right)$ is

(1) 91

(2) 12

(3) 36

(4) 125

Sol. Answer (4)

$$\text{Let } f(\theta) = 64 \sec \theta + 27 \operatorname{cosec} \theta$$

$$\Rightarrow f'(\theta) = 64 \sec \theta \tan \theta - 27 \operatorname{cosec} \theta \cot \theta = \frac{64 \sin^3 \theta - 27 \cos^3 \theta}{\sin \theta \cos \theta}$$

For maxima–minima of $f(\theta)$

$$f'(\theta) = 0 \Rightarrow \tan \theta = \frac{3}{4}$$

$$\text{Since } f''(\theta) > 0, f(\theta) \text{ has minimum value for } \tan \theta = \frac{3}{4}$$

$$\text{and Min } f(\theta) = 125$$

85. For the function $f(x) = x^2 - 6x + 8$, $2 \leq x \leq 4$, the value of x for which $f'(x)$ vanishes is

(1) 3

(2) $\frac{5}{2}$

(3) $\frac{9}{4}$

(4) $\frac{7}{2}$

Sol. Answer (1)

$$f'(x) = 2x - 6$$

$$f'(x) = 0$$

$$\text{For } x = 3$$

[Miscellaneous]

86. Which of the following statement is true for the function?

$$f(x) = \begin{cases} \sqrt{x} & , x \geq 1 \\ x^3 & , 0 \leq x < 1 \\ \frac{x^3}{3} - 4x & , x < 0 \end{cases}$$

- (1) It is monotonic increasing $\forall x \in R$
- (2) $f'(x)$ fails to exist for 3 distinct real values of x
- (3) $f'(x)$ changes its sign twice as x varies from $(-\infty, \infty)$
- (4) Function attains its extreme values at x_1 and x_2 , such that $x_1 x_2 > 0$

Sol. Answer (3)

$$f(x) = \begin{cases} \sqrt{x} & x \geq 1 \\ x^3 & 0 \leq x \leq 1 \\ \frac{x^3}{3} - 4x & x < 0 \end{cases}$$

$$f'(x) = \begin{cases} \frac{1}{2\sqrt{x}} & x \geq 1 \\ 3x^2 & 0 \leq x \leq 1 \\ x^2 - 4 & x < 0 \end{cases}$$

$f'(x)$ changes its sign two times at $x = 0, -2$

87. Consider the function $f(x) = x \cos x - \sin x$, then identify the statement which is correct.

- (1) f is neither odd nor even
- (2) f is monotonic increasing in $\left(0, \frac{\pi}{2}\right)$
- (3) f has a maxima at $x = -\pi$
- (4) f has a minima at $x = -\pi$

Sol. Answer (3)

$f(x)$ is odd function

$$f'(x) = -x \sin x + \cos x - \cos x$$

$$f'(x) = -x \sin x$$

$$f''(x) = -x \cos x - \sin x$$

at $x = -\pi$

$f''(x) < 0$ point of maxima

88. For the cubic, $f(x) = 2x^3 + 9x^2 + 12x + 1$, which one of the following statement, does not hold good?

- (1) $f(x)$ is non-monotonic
- (2) Increasing in $(-\infty, -2) \cup (-1, \infty)$ and decreasing in $(-2, -1)$
- (3) $f : R \rightarrow R$ is bijective
- (4) Inflection point occurs at $x = -\frac{3}{2}$

Sol. Answer (3)

$$f'(x) = 6x^2 + 18x + 12$$

$$= 6(x^2 + 3x + 2)$$

$$= 6(x + 1)(x + 2),$$

'f' is increasing in $(-\infty, -2) \cup (-1, \infty)$ and decreasing in $(-2, -1)$.

$$f''(x) = 6(2x + 3)$$

$$f''(x) \text{ changes sign at } x = -\frac{3}{2}$$

$$\therefore \text{ Point of inflection occur at } x = -\frac{3}{2}$$

89. Let $h(x)$ be a twice continuously differentiable positive function on an open interval J . Let

$$g(x) = \ln(h(x)) \text{ for each } x \in J.$$

Suppose $(h'(x))^2 > (h''(x))h(x)$ for each $x \in J$ then

(1) g is increasing on J

(2) g is decreasing on J

(3) g is concave up on J

(4) $g'(x)$ is decreasing function

Sol. Answer (4)

$$g(x) = \ln(h(x))$$

$$g'(x) = \frac{1}{h(x)} \cdot h'(x)$$

$$g''(x) = \frac{h(x) \cdot h''(x) - (h'(x))^2}{h^2(x)}$$

$$g''(x) < 0$$

So, $g'(x)$ is decreasing function and g is concave down on J .

90. If $1^\circ = \alpha$ radians then the approximate value of $\cos 60^\circ 1'$ is

$$(1) \frac{1}{2} + \frac{\alpha\sqrt{3}}{120}$$

$$(2) \frac{1}{2} - \frac{\alpha}{120}$$

$$(3) \frac{1}{2} - \frac{\alpha\sqrt{3}}{120}$$

$$(4) \frac{1}{2}$$

Sol. Answer (3)

$$\text{Let } f(x) = \cos x \Rightarrow f'(x) = -\sin x$$

$$f(x + h) = f(x) + f'(x)\Delta y$$

$$\cos 60^\circ 1' = \frac{1}{2} - \frac{\sqrt{3}}{2} 1' = \frac{1}{2} - \frac{\sqrt{3}}{2} \frac{\alpha}{60} = \frac{1}{2} - \frac{\sqrt{3}\alpha}{120}$$

91. Let the equation $x - \sin x = a$ has a unique root in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, then

(1) $a \in \left[-\frac{\pi}{2} + 1, \infty\right)$

(2) $a \in \left(-\infty, \frac{\pi}{2} - 1\right]$

(3) $a \in \left[1 - \frac{\pi}{2}, \frac{\pi}{2} - 1\right]$

(4) $a \in \mathbb{R} - \left(1 - \frac{\pi}{2}, \frac{\pi}{2} - 1\right)$

Sol. Answer (3)

$$\sin x = x - a$$

Case-1

If $a > 0$

If $y = x - a$ touch $\sin x$ at $x = \frac{\pi}{2}$

$$\Rightarrow a = \frac{\pi}{2} - 1$$

Case-2

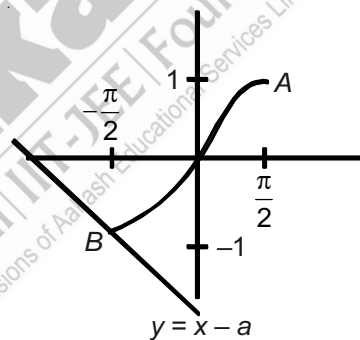
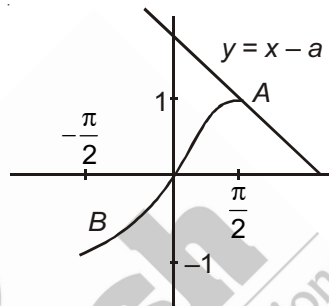
If $a < 0$

If $y = x - a$ touch

$$y = \sin x \text{ at } x = -\frac{\pi}{2}$$

$$\Rightarrow a = 1 - \frac{\pi}{2}$$

$$\text{Hence } a \in \left[1 - \frac{\pi}{2}, \frac{\pi}{2} - 1\right].$$



92. Number of real roots of the equation

$$e^{x-1} - x = 0 \text{ is}$$

(1) 1

(2) 2

(3) 3

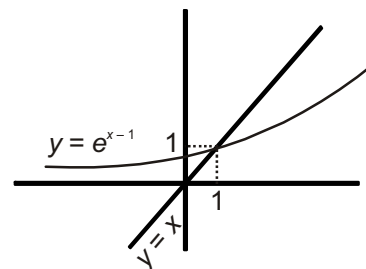
(4) 0

Sol. Answer (1)

The number of real roots of the equation $e^{x-1} = x$ is the number of points of intersection of the graph of the two curves $y = e^{x-1}$ and $y = x$.

$$e^{x-1} = x$$

From graph it is clear that the number of real root = 1.



93. The number of critical points of the function

$$f(x) = \int_0^x e^t (t-1)(t-2)(t-3) dt \text{ is}$$

(1) 3

(2) 4

(3) 12

(4) 2

Sol. Answer (1)

$$f(x) = \int_0^x e^t (t-1)(t-2)(t-3) dt$$

$$\Rightarrow f'(x) = e^x (x-1)(x-2)(x-3)$$

For critical points of $f(x)$, we must have $f'(x) = 0 = e^x (x-1)(x-2)(x-3)$

$$\Rightarrow x = 1, 2, 3$$

$\therefore$ Number of critical point = 3.

94. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbb{R}$, and $f\left(\frac{1}{2}\right) = \frac{1}{2}$, $f(1) = 1$, then

[JEE(Advanced)-2017]

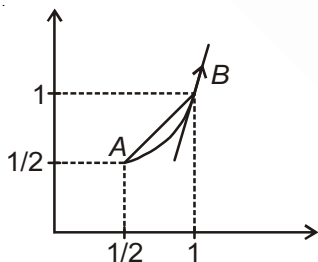
(1) $\frac{1}{2} < f'(1) \leq 1$

(2) $0 < f'(1) \leq \frac{1}{2}$

(3) $f'(1) \leq 0$

(4) $f'(1) > 1$

Sol. Answer (4)



$$f''(x) > 0, f\left(\frac{1}{2}\right) = \frac{1}{2} \text{ and } f(1) = 1$$

$f'(x)$ is always increasing

$$f'(1) > \frac{f(1) - f\left(\frac{1}{2}\right)}{1 - \frac{1}{2}}$$

$$f'(1) > 1$$

Slope of tangent at B > Slope of chord AB .

SECTION - B

Objective Type Questions (More than one options are correct)

1. If the line, $ax + by + c = 0$ is normal to the curve $xy = 2$, then
- (1) $a < 0, b > 0$ (2) $a > 0, b < 0$
 (3) $a > 0, b > 0$ (4) $a < 0, b < 0$

Sol. Answer (1, 2)

$$\text{Slope of the normal } (ax + by + c = 0) = -\frac{a}{b}$$

The given curve is

$$xy = 2$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y}{x} = \frac{b}{a} \Rightarrow \frac{b}{a} < 0 \Rightarrow a < 0, b > 0 \text{ or } a > 0, b < 0$$

2. The equation of the tangents to the curve $y = x^4$ from the point $(2, 0)$, are given by

- (1) $y = 0$ (2) $y - 1 = 5(x - 1)$
 (3) $y - \frac{4096}{81} = \frac{2048}{27} \left(x - \frac{8}{3} \right)$ (4) $y - \frac{32}{243} = \frac{80}{81} \left(x - \frac{2}{3} \right)$

Sol. Answer (1, 3)

Let $P(x_1, y_1)$ be a point on the curve

$$y = x^4$$

$$\Rightarrow \frac{dy}{dx} = 4x^3$$

$$m = \left(\frac{dy}{dx} \right)_{(x_1, y_1)} = 4x_1^3$$

The equation of tangent to the given curve at (x_1, y_1) is given by

$$y - y_1 = 4x_1^3(x - x_1)$$

Which passes through $(2, 0)$ hence

$$-y_1 = 4x_1^3(2 - x_1)$$

$$\Rightarrow 8x_1^3 - 4x_1^4 + y_1 = 0$$

$$4x_1^4 - 8x_1^3 - y_1 = 0 \quad \dots(1)$$

$$(x_1, y_1) \text{ lie on } y = x^4 \Rightarrow y_1 = x_1^4$$

$$\Rightarrow x_1 = 0, \frac{8}{3}$$

$$\Rightarrow 4x_1^4 - 8x_1^3 - x_1^4 = 0$$

$$\Rightarrow y_1 = 0, \frac{4096}{81} \quad m = 0 \text{ and } \frac{2048}{27}$$

Thus the equations of tangents are

$$y = 0 \text{ and } y - \frac{4096}{81} = \frac{2048}{27} \left(x - \frac{8}{3} \right)$$

3. Suppose $f'(x)$ exists for each x and

$$h(x) = f(x) - (f(x))^2 + (f(x))^3 \quad \forall x \in R, \text{ then}$$

(1) h is increasing whenever f is increasing

(2) h is increasing whenever f is decreasing

(3) h is decreasing whenever f is decreasing

(4) Nothing can be said in general

Sol. Answer (1, 3)

$$\text{We have, } h(x) = f(x) - (f(x))^2 + f((x))^3$$

$$\Rightarrow h'(x) = f'(x) - 2f(x)f'(x) + 3(f(x))^2 f'(x) = f'(x)[3(f(x))^2 - 2f(x) + 1]$$

Since $3(f(x))^2 - 2f(x) + 1 > 0$ for real values of $f(x)$ as $4 - 12 < 0$

$$\Rightarrow h'(x) > 0 \text{ if } f'(x) > 0 \text{ and } h'(x) < 0 \text{ if } f'(x) < 0$$

Hence $h(x)$ is increasing or decreasing function according as $f(x)$ increases or decreases.

4. Let $f'(x) = g(x)(x - a)^2$, where $g(a) \neq 0$ and g is continuous at $x = a$ then

(1) f is increasing near a if $g(a) < 0$

(2) f is decreasing near a if $g(a) > 0$

(3) f is decreasing near a if $g(a) < 0$

(4) f is increasing near a if $g(a) > 0$

Sol. Answer (3, 4)

$$f'(x) = g(x)(x - a)^2$$

In the neighbourhood of $x = a$, we have $x = a \pm h$ as $g(x)$ is continuous at $x = a$

$$\Rightarrow f'(a) = g(a)h^2$$

$$\text{if } g(a) > 0 \Rightarrow f'(a) > 0$$

$$\text{if } g(a) < 0 \Rightarrow f'(a) < 0.$$

$f(x)$ is increasing in the neighbourhood of a if $g(a) > 0$ and is decreasing if $g(a) < 0$

5. Let $f(x) = (x^2 - 1)^n (x^2 + x - 1)$ then $f(x)$ has local minimum at $x = 1$, when

(1) $n = 2$

(2) $n = 3$

(3) $n = 4$

(4) $n = 5$

Sol. Answer (1, 3)

$$f(x) = (x^2 - 1)^n (x^2 + x - 1)$$

$f(x)$ has local extremum at $x = 1$

then either

$$f(1) < f(1-h) \text{ and } f(1) < f(1+h) \text{ or } f(1) > f(1-h) \text{ and } f(1) > f(1+h)$$

where $f(1) = 0$. Thus for local minimum at $x = 1$

$$f(1-h) > 0 \text{ and } f(1+h) > 0 \text{ or } f(1-h) < 0 \text{ and } f(1+h) < 0$$

If $f(1-h) > 0$ and $f(1+h) > 0 \forall h > 0$

$$(-h)^n(2-h)^n(h^2-3h+1) \text{ and } (h)^n(2+h)^n(h^2+3h+1) > 0$$

$$\text{i.e., } (-h)^n > 0$$

$n \in$ even number

Hence $n = 2, 4$.

6. Which of the following is correct?

$$(1) e^2 > 2^e$$

$$(2) e^3 > 3^e$$

$$(3) e^\pi > \pi^e$$

$$(4) \sqrt{e}^\pi > \left(\frac{\pi}{2}\right)^e$$

Sol. Answer (1, 2, 3, 4)

Consider the function

$$f(x) = x^{1/x} = e^{\frac{1}{x} \ln x}$$

$$f'(x) = x^{1/x} \left[\frac{1}{x^2} - \frac{1}{x^2} \ln x \right] = \frac{x^{1/x}}{x^2} [1 - \ln x]$$

if $x < e \Rightarrow f'(x) > 0$ i.e., $f(x)$ is increasing in $(0, e)$

if $x > e \Rightarrow f'(x) < 0$ i.e., $f(x)$ is decreasing in (e, ∞)

Now, $2 < e < 3$

$$\Rightarrow f(e) > f(2) \Rightarrow e^{1/e} > 2^{1/2}$$

$$\Rightarrow e^2 > 2^e \text{ option (1) is correct}$$

Also $e < 3$ and $f(x)$ decreases in (e, ∞)

$$\Rightarrow f(e) > f(3) \Rightarrow e^3 > 3^e \text{ option (2) is correct}$$

$$\pi > e$$

$$\Rightarrow f(\pi) < f(e) \Rightarrow e^{1/\pi} > e^{1/e}$$

$$\Rightarrow e^\pi > \pi^e \text{ option (3) is correct}$$

Moreover

$$\frac{\pi}{2} < e \Rightarrow f\left(\frac{\pi}{2}\right) < f(e) \Rightarrow \left(\frac{\pi}{2}\right)^{\frac{1}{\pi/2}} < e^{1/e}$$

$$e^{\pi/2} > \left(\frac{\pi}{2}\right)^e$$

$$\sqrt{e}^\pi > \left(\frac{\pi}{2}\right)^e \text{ option (4) is correct.}$$

7. An extremum value of the function $f(x) = (\sin^{-1} x)^3 + (\cos^{-1} x)^3$ ($-1 \leq x \leq 1$) is

(1) $\frac{7\pi^3}{8}$

(2) $\frac{\pi^3}{8}$

(3) $\frac{\pi^3}{32}$

(4) $\frac{\pi^3}{16}$

Sol. Answer (1, 3)

Let $\sin^{-1} x = y$

$$\cos^{-1} x = \frac{\pi}{2} - y$$

$$f(y) = y^3 + \left(\frac{\pi}{2} - y\right)^3$$

$$f'(y) = 3y^2 - 3\left(\frac{\pi}{2} - y\right)^2 \Rightarrow y = \frac{\pi}{4}$$

$\frac{\pi}{4}$ is point of minima

$$\therefore \min = \frac{\pi^3}{32}$$

$$\max = \frac{7\pi^3}{8} \text{ at } y = \frac{-\pi}{2}$$

8. Let the function $f(x) = \int_{-2}^x t(e^t - 1)(t+1)(t-2)(t-3)dt$ has a point of minima at x equal to

(1) 0

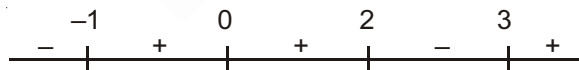
(2) -1

(3) 2

(4) 3

Sol. Answer (2, 4)

$$f'(x) = x(e^x - 1)(x+1)(x-2)(x-3)$$



9. Let $f(x) = |\sin x + \cos x|$ then for 'r'

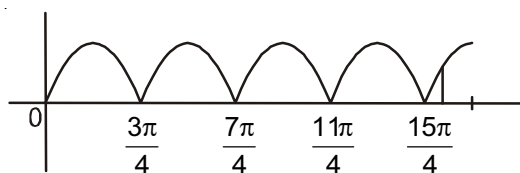
(1) Point of relative maxima in $[0, 4\pi]$ is 4

(2) Point of relative maxima in $[0, 4\pi]$ is 7

(3) Point of relative minima in $(0, 4\pi)$ is 4

(4) Point of relative minima in $(0, 4\pi)$ is 8

Sol. Answer (1, 3)



10. For every pair of continuous functions $f, g : [0, 1] \rightarrow \mathbb{R}$ such that $\max \{f(x) : x \in [0, 1]\} = \max \{g(x) : x \in [0, 1]\}$, the correct statement(s) is(are) [JEE(Advanced)-2014]

- (1) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
- (2) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
- (3) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0, 1]$
- (4) $(f(c))^2 = (g(c))^2$ for some $c \in [0, 1]$

Sol. Answer (1, 4)

Let f and g be maximum at c_1 and c_2 respectively, $c_1, c_2 \in (0, 1)$.

Let $h(x) = f(x) - g(x)$

Now $h(c_1) = f(c_1) - g(c_1) = +ve$

and $h(c_2) = f(c_2) - g(c_2) = -ve$

$\therefore h(x) = 0$ has at least one root in (c_1, c_2)

$\therefore f(x) = g(x)$ for some $x = c \in (c_1, c_2)$

$\therefore f(c) = g(c)$ for some $c \in (0, 1)$

Clearly (1, 4) are correct.

11. Let $a \in \mathbb{R}$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by

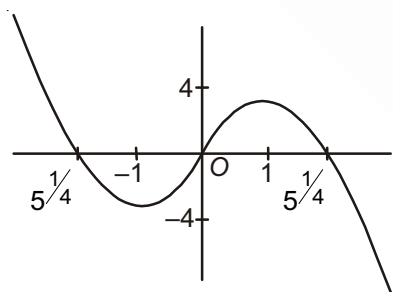
$$f(x) = x^5 - 5x + a.$$

Then

- (1) $f(x)$ has three real roots if $a > 4$
- (2) $f(x)$ has only one real root if $a > 4$
- (3) $f(x)$ has three real roots if $a < -4$
- (4) $f(x)$ has three real roots if $-4 < a < 4$

[JEE(Advanced)-2014]

Sol. Answer (2, 4)



$$f(x) = x^5 - 5x + a$$

$$\text{if } f(x) = 0 \Rightarrow a = 5x - x^5 = g(x)$$

$$g(x) = 0 \Rightarrow x = 0, 5^{1/4}, -5^{1/4}$$

$$g'(x) = 0 \Rightarrow 5 - 5x^4 = 0 \Rightarrow x = 1, -1$$

$$g(-1) = -4$$

$$g(1) = 4$$

If $a \in (-4, 4) \Rightarrow f(x) = 0$ has 3 real roots

if $a > 4$ or $a < -4 \Rightarrow f(x) = 0$ has only 1 real root.

12. Let $f, g : [-1, 2] \rightarrow \mathbb{R}$ be continuous functions which are twice differentiable on the interval $(-1, 2)$. Let the values of f and g at the points $-1, 0$ and 2 be as given in the following table:

	$x = -1$	$x = 0$	$x = 2$
$f(x)$	3	6	0
$g(x)$	0	1	-1

In each of the intervals $(-1, 0)$ and $(0, 2)$ the function $(f - 3g)''$ never vanishes. Then the correct statement(s) is(are) [JEE(Advanced)-2015]

- (1) $f'(x) - 3g'(x) = 0$ has exactly three solutions in $(-1, 0) \cup (0, 2)$
- (2) $f'(x) - 3g'(x) = 0$ has exactly one solution in $(-1, 0)$
- (3) $f'(x) - 3g'(x) = 0$ has exactly one solution in $(0, 2)$
- (4) $f'(x) - 3g'(x) = 0$ has exactly two solutions in $(-1, 0)$ and exactly two solutions in $(0, 2)$

Sol. Answer (2, 3)

$$h(x) = f(x) - 3g(x)$$

$$h(-1) = 3 - 0 = 3$$

$$h(0) = 6 - 3 = 3$$

$$h(2) = 0 + 3 = 3$$

$h'(x)$ has atleast one root in $(-1, 0)$

and $h'(x)$ has atleast one in $(0, 2)$

hence $h''(x)$ atleast one root in $(-1, 2)$ but

Also, $(f - 3g)'' \neq 0$ in $(-1, 0)$ and $(0, 2)$

$h''(x) = 0$ only at $x = 0$, so

$f'(x) - 3g'(x)$ has exactly one solution in $(-1, 0)$ and $(0, 2)$

13. If $f(x) = \begin{vmatrix} \cos(2x) & \cos(2x) & \sin(2x) \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}$, then

[JEE(Advanced)-2017]

- (1) $f(x)$ attains its maximum at $x = 0$
- (2) $f(x)$ attains its minimum at $x = 0$
- (3) $f'(x) = 0$ at more than three points in $(-\pi, \pi)$
- (4) $f'(x) = 0$ at exactly three points in $(-\pi, \pi)$

Sol. Answer (1, 3)

$$C_1 \rightarrow C_1 - C_2$$

$$f(x) = \begin{vmatrix} 0 & \cos 2x & \sin 2x \\ -2\cos x & \cos x & -\sin x \\ 0 & \sin x & \cos x \end{vmatrix}$$

$$\Rightarrow f(x) = 2\cos x(\cos 2x \cos x - \sin x \sin 2x)$$

$$\Rightarrow f(x) = 2\cos x \cos 3x; (f(0) = 2 \text{ maximum at } x = 0)$$

$$\Rightarrow f(x) = \cos 4x + \cos 2x$$

$$f'(x) = -2\sin 2x(4\cos 2x + 1)$$

$$\Rightarrow \sin 2x = 0 \text{ or } \cos 2x = -\frac{1}{4}$$

$$2x = 0, \pi, -\pi$$

$$x = 0, \frac{\pi}{2}, -\frac{\pi}{2} \text{ and } \cos 2x = -\frac{1}{4} \text{ gives 4 solutions in } (-\pi, \pi)$$

$$\therefore \text{Total number of solutions} = 7$$

14. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f'(x) > 2f(x)$ for all $x \in \mathbb{R}$, and $f(0) = 1$, then

[JEE(Advanced)-2017]

(1) $f(x)$ is increasing in $(0, \infty)$

(2) $f'(x) < e^{2x}$ in $(0, \infty)$

(3) $f(x) > e^{2x}$ in $(0, \infty)$

(4) $f(x)$ is decreasing in $(0, \infty)$

Sol. Answer (1, 3)

$$f'(x) - 2f(x) > 0$$

$$e^{-2x} \cdot f'(x) - 2e^{-2x} \cdot f(x) > 0$$

$$\frac{d}{dx}(e^{-2x}f(x)) > 0 \Rightarrow e^{-2x} \cdot f(x) \text{ is increasing function.}$$

$$e^{-2x} \cdot f(x) > 1 \text{ for all } x \in (0, \infty)$$

$$f(x) > e^{2x}$$

$$\therefore f'(x) > 2f(x) > e^{2x} > 0$$

$$\therefore f(x) \text{ is increasing}$$

$$\text{Also as, } f'(x) = \frac{f(x) - f(0)}{x - 0} \Rightarrow f'(x) = \frac{f(x) - 1}{x}$$

$$\text{i.e., } f'(x) > e^{2x} \forall x \in (0, 1)$$

$$< e^{2x} \forall x \in (1, \infty)$$

15. For every twice differentiable function $f : \mathbb{R} \rightarrow [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 85$, which of the following statement(s) is (are) TRUE? [JEE(Advanced)-2018]

- (1) There exist $r, s \in \mathbb{R}$, where $r < s$, such that f is one-one on the open interval (r, s)
 (2) There exists $x_0 \in (-4, 0)$ such that $|f'(x_0)| \leq 1$
 (3) $\lim_{x \rightarrow \infty} f(x) = 1$
 (4) There exists $\alpha \in (-4, 4)$ such that $f(\alpha) + f''(\alpha) = 0$ and $f'(\alpha) \neq 0$

Sol. Answer (1, 2, 4)

$$(f(0))^2 + (f'(0))^2 = 85$$

$$\text{If } f(x) = \text{constant} \Rightarrow f'(x) = 0$$

$$\Rightarrow f(0) = \pm \sqrt{85} \text{ but } f(x) \in [-2, 2] \Rightarrow f(0) \neq \pm \sqrt{85}$$

$\therefore f(x)$ is continuous function which is not constant

$\therefore$ It is always possible to find (r, s) where $f(x)$ is one one.

$$\text{Let } f(x) = a \sin bx$$

$$f'(x) = ab \cos bx$$

$$\therefore f(x) \in [-2, 2] \Rightarrow a \in [-2, 2]$$

$$\therefore (f(0))^2 + (f'(0))^2 = 85$$

$$\therefore a^2 b^2 = 85$$

$$\text{For this function } \lim_{x \rightarrow \infty} f(x) \neq 1$$

$$\text{Using LMVT, } |f'(x)| = \left| \frac{f(-4) - f(0)}{-4} \right| \leq 1, \text{ as } f(x) \in [-2, 2]$$

$$\text{Also } f(\alpha) + f''(\alpha) = a \sin b\alpha \cdot (1 - b^2) = 0 \text{ is true for some } \alpha \in (-4, 4)$$

SECTION - C

Linked Comprehension Type Questions

Comprehension I

Let the solution set of the inequation $\left(\sin x - \frac{1}{2}\right) \left(\sin x - \frac{1}{\sqrt{2}}\right) \leq 0$ in $\left[\frac{\pi}{2}, \pi\right]$ be A and let solution set of equation $\sin^{-1}(3x - 4x^3) = 3 \sin^{-1} x$ be B . Now define a function $f : A \rightarrow B$.

1. There are exactly two linear onto functions that can be drawn from A to B , one of them is

- (1) $\frac{12x}{\pi} - \frac{19}{2}$ (2) $-\frac{12x}{\pi} - \frac{19}{2}$ (3) $\frac{12x}{\pi} + \frac{19}{2}$ (4) $12\pi + \frac{19}{2}x$

2. Among the following statements which is not true?
- (1) $\cos\theta = f(x)$, for some $x \in A$, $\theta \in R$ (2) $\sin\theta + \cos\theta = f(x)$, for some $x \in A$, $\theta \in R$
 (3) $\tan\theta = f(x)$ for some $x \in A$, $\theta \in R$ (4) $\sec\theta = f(x)$ for some $x \in A$ and $\theta \in R$
3. If $f: A \xrightarrow{\text{on to}} B$ is a strictly decreasing function, then ordered pair (α, β) satisfying the equation

$$\sqrt{\beta^2 - 6\beta + \frac{37}{4}} = f(\alpha) \text{ is}$$

- (1) $\left(\frac{3\pi}{4}, \frac{1}{2}\right)$ (2) $\left(\frac{5\pi}{6}, 3\right)$ (3) $\left(\frac{3\pi}{4}, 3\right)$ (4) $(3, \infty)$

Solutions of Comprehension I

$$\left(\sin x - \frac{1}{2}\right)\left(\sin x - \frac{1}{\sqrt{2}}\right) \leq 0$$

$$\Rightarrow x \in \left[\frac{3\pi}{4}, \frac{5\pi}{6}\right] = A$$

$$\text{and } \sin^{-1}(3x - 4x^3) = 3\sin^{-1}(x)$$

$$\Rightarrow x \in \left[-\frac{1}{2}, \frac{1}{2}\right] = B$$

$$f: A \rightarrow B$$

1. Answer (1)

$$\text{Let } f(x) = ax + b$$

$$\text{Case - 1 : } f\left(\frac{3\pi}{4}\right) = a\frac{3\pi}{4} + b \text{ and } f\left(\frac{5\pi}{6}\right) = a\frac{5\pi}{6} + b$$

$$-\frac{1}{2} = \frac{3\pi}{4}a + b \quad \frac{1}{2} = \frac{5\pi}{6}a + b$$

$$\text{On solving } a = \frac{12}{\pi}, b = -\frac{19}{2}$$

$$\text{i.e., } f(x) = \frac{12}{\pi}x - \frac{19}{2}$$

$$\text{Case - 2 : } f\left(\frac{3\pi}{4}\right) = \frac{3\pi}{4}a + b \text{ and } f\left(\frac{5\pi}{6}\right) = \frac{5\pi}{6}a + b$$

$$\Rightarrow \frac{1}{2} = \frac{3\pi}{4}a + b \quad -\frac{1}{2} = \frac{5\pi}{6}a + b$$

On solving

$$a = -\frac{12}{\pi}, b = \frac{19}{2}$$

$$f(x) = -\frac{12}{\pi}x + \frac{19}{2}$$

2. Answer (4)

3. Answer (3)

As, $f : A \xrightarrow{\text{onto}} B$ is strictly decreasing function

$$\text{then, } f : \left[\frac{3\pi}{4}, \frac{5\pi}{6} \right] \rightarrow \left[-\frac{1}{2}, \frac{1}{2} \right]$$

$$\text{then, } \frac{3\pi}{4} < x < \frac{5\pi}{6} \Rightarrow f\left(\frac{3\pi}{4}\right) > f(x) > f\left(\frac{5\pi}{6}\right)$$

$$\text{Now, } \sqrt{\beta^2 - 6\beta + \frac{37}{4}} = f(\alpha) \Rightarrow \sqrt{(\beta - 3)^2 + \frac{1}{4}} = f(\alpha)$$

$$\text{If } \beta = 3 \Rightarrow f(\alpha) = \frac{1}{2} \text{ i.e., } \alpha = \frac{3\pi}{4}$$

$$\text{i.e., ordered pair } (\alpha, \beta) \text{ is } \left(\frac{3\pi}{4}, 3 \right).$$

Comprehension II

When the product of two positive numbers is constant, then the minimum value of their sum is 2 times the square root of their product. Also when the sum of two positive numbers is constant, the maximum value of their product is $\frac{1}{4}$ time the square of their sum. We get these minimum and maximum values only when the two numbers are equal.

1. The minimum value of $f(x) = \frac{1}{2\{-x\}} + \{-x\}$, (where $\{x\}$ is fractional part of x) is

(1) $\sqrt{2}$

(2) $\sqrt{2} + 1$

(3) $2\sqrt{2}$

(4) $\sqrt{2} - 1$

Sol. Answer (1)

$$\frac{1}{2\{-x\}} + \{-x\} \geq 2\sqrt{\frac{\{-x\}}{2\{-x\}}} \geq \sqrt{2}$$

2. The range of $g(x) = (\cot^{-1}x)(\cot^{-1}(-x))$ is

(1) $\left(\frac{\pi^2}{6}, \frac{\pi^2}{4} \right)$

(2) $\left(0, \frac{\pi^2}{4} \right)$

(3) $\left(0, \frac{\pi^2}{6} \right)$

(4) $(0, \pi^2)$

Sol. Answer (2)

$$g(x) = \cot^{-1}x(\pi - \cot^{-1}x) = \pi \cot^{-1}x - (\cot^{-1}x)^2 = -\left(\cot^{-1}x - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{4}$$

$$\text{Completing the squares } \left(0, \frac{\pi^2}{4} \right)$$

3. The range of $f(x) = \ln \left[\frac{x}{4e^3} + \frac{e}{x} \right]$ is

(1) $[0, \infty)$

(2) $[-1, \infty)$

(3) $(-\infty, 0]$

(4) $(0, \infty)$

Sol. Answer (2)

$$f(x) = \ln \left[\frac{x}{4e^3} + \frac{e}{x} \right]$$

$$\therefore \left(\ln \left(\frac{1}{e} \right), \ln \infty \right) \quad [-1, \infty)$$

$$\frac{x}{4e^3} + \frac{e}{x} \geq 2\sqrt{\frac{1}{4e^2}} > \frac{1}{e}$$

Comprehension III

Among several applications of maxima and minima is finding the largest term of a sequence. Let $\langle a_n \rangle$ be a sequence. Consider $f(x)$ obtained by replacing x by n e.g., let $a_n = \frac{n}{n+1}$ consider $f(x) = \frac{x}{x+1}$ on $[1, \infty)$

$$f'(x) = \frac{1}{(1+x)^2} > 0 \quad \forall x \in R.$$

$$\text{Hence max. } f(x) = \lim_{x \rightarrow \infty} f(x) = 1$$

1. The largest term of the sequence $a_n = \frac{n^2}{n^3 + 200}$ is

(1) $\frac{29}{453}$

(2) $\frac{49}{543}$

(3) $\frac{43}{543}$

(4) $\frac{41}{451}$

Sol. Answer (2)

$$f(x) = \frac{x^2}{x^3 + 200}$$

$$\Rightarrow f'(x) = \frac{x(400 - x^3)}{(x^3 + 200)^2}$$

$$f'(x) > 0 \quad \text{if } 0 < x < (400)^{1/3}$$

$$f'(x) < 0 \quad \text{if } x > (400)^{1/3}$$

$$\text{i.e., } f(x) \text{ is maximum at } x = (400)^{1/3}$$

Since $7 < (400)^{1/3} < 8$

Hence max term of the given sequence

$$a_n = \frac{n^2}{n^3 + 200} \text{ is } a_7 = \frac{49}{543}$$

2. The largest term of the sequence $a_n = \frac{n}{n^2 + 10}$ is

(1) $\frac{3}{19}$

(2) $\frac{2}{13}$

(3) 1

(4) $\frac{1}{7}$

Sol. Answer (1)

Let $a_n = \frac{n}{n^2 + 10}$

Let $f(x) = \frac{x}{x^2 + 10}$

$$\Rightarrow f'(x) = \frac{x^2 + 10 - 2x^2}{(x^2 + 10)^2} = \frac{10 - x^2}{(x^2 + 10)^2} = \frac{-(x - \sqrt{10})(x + \sqrt{10})}{(x^2 + 10)^2}$$

$$\Rightarrow f'(x) > 0 \text{ if } x < \sqrt{10}$$

$$\text{and } f'(x) < 0 \text{ if } x > \sqrt{10}$$

$$\text{i.e., } f(x) \text{ is max at } x = \sqrt{10}$$

$$\text{since } 3 < \sqrt{10} < 4$$

$$\text{and } a_3 = \frac{9}{19} > \frac{4}{26} = a_4$$

Hence largest term of the sequence

$$a_n = \frac{n}{n^2 + 10} \text{ is } a_3 = \frac{3}{19}$$

3. If $f(x)$ is the function required to find largest term in question 1 then

(1) $f(x)$ increase $\forall x$

(2) $f(x)$ decrease $\forall x$

(3) $f(x)$ has a maxima at $x = (400)^{1/3}$

(4) $f(x)$ increases on $[0, 9]$

Sol. Answer (3)

From (1), $f(x)$ has maxima at $x = (400)^{1/3}$

Comprehension IV

Let $f(x) = x + \log_e x - x \log_e x, x \in (0, \infty)$

- Column 1 contains information about zeros of $f(x), f'(x)$ and $f''(x)$.
- Column 2 contains information about the limiting behaviour of $f(x), f'(x)$ and $f''(x)$ at infinity.
- Column 3 contains information about increasing/decreasing nature of $f(x)$ and $f'(x)$.

Column 1	Column 2	Column 3
(I) $f(x) = 0$ for some $x \in (1, e^2)$	(i) $\lim_{x \rightarrow \infty} f(x) = 0$	(P) f is increasing in $(0, 1)$
(II) $f'(x) = 0$ for some $x \in (1, e)$	(ii) $\lim_{x \rightarrow \infty} f(x) = -\infty$	(Q) f is decreasing in (e, e^2)
(III) $f'(x) = 0$ for some $x \in (0, 1)$	(iii) $\lim_{x \rightarrow \infty} f'(x) = -\infty$	(R) f' is increasing in $(0, 1)$
(IV) $f''(x) = 0$ for some $x \in (1, e)$	(iv) $\lim_{x \rightarrow \infty} f''(x) = 0$	(S) f' is decreasing in (e, e^2)

[JEE(Advanced)-2017]

- Which of the following options is the only correct combination?
 (1) (III) (iv) (P) (2) (I) (ii) (R) (3) (II) (iii) (S) (4) (IV) (i) (S)
 - Which of the following options is the only incorrect combination?
 (1) (II) (iv) (Q) (2) (III) (i) (R) (3) (I) (iii) (P) (4) (II) (iii) (P)
 - Which of the following options is the only correct combination?
 (1) (I) (i) (P) (2) (II) (ii) (Q) (3) (IV) (iv) (S) (4) (III) (iii) (R)
- Answer (3)
 - Answer (2)
 - Answer (2)

Solution of Comprehension IV

$$f(x) = x + \log_e x - x \log_e x$$

$$(I) \quad f(1) = 1 + \log 1 - 1 \log 1 = 1 > 0$$

$$f(e^2) = e^2 + \log_e e^2 - e^2 \log_e e^2 = e^2 + 2 - 2e^2 = 2 - e^2 < 0$$

$\Rightarrow$ (I) is true.

$$(II) \quad f'(x) = 1 + \frac{1}{x} - \log_e x - 1$$

$$f'(1) = 1$$

$$f'(e) = \frac{1}{e} - 1 < 0$$

$\Rightarrow$ (II) is true.

$$(III) f''(x) = -\frac{1}{x^2} - \frac{1}{x} < 0 \text{ for all } x \in (0, \infty)$$

So, $f'(x)$ is decreasing

So, min. of $f'(x)$ in $(0, 1)$ is $f'(1) = 1$

So, (III) is false.

$$(IV) f''(1) = -2 < 0$$

$$f''(e) = -\frac{1}{e^2} - \frac{1}{e} < 0$$

So, (IV) is false.

Column 2

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} x + \log_e x - x \log_e x = \lim_{t \rightarrow 0^+} \frac{1}{t} - \log_e t + \frac{\log_e t}{t}$$

$$\lim_{t \rightarrow 0^+} \frac{1 - t \log_e t + \log_e t}{t} = -\infty$$

(i) is false

(ii) is true

$$(iii) \lim_{x \rightarrow \infty} f'(x) = \lim_{x \rightarrow \infty} \frac{1}{x} - \log_e x = \lim_{t \rightarrow 0^+} t + \log_e t = -\infty \text{ (True)}$$

$$(iv) \lim_{x \rightarrow \infty} f''(x) = -\frac{1}{x^2} - \frac{1}{x} = \lim_{x \rightarrow \infty} \frac{-1-x}{x^2} = 0 \text{ (True)}$$

Column 3

$$(P) f'(x) = \frac{1}{x} - \log_e x > 0 \text{ is true}$$

$$\text{for } x \in (0, 1), \frac{1}{x} \in (1, \infty)$$

$$\log_e x \in (-\infty, 0)$$

$$(Q) f'(x) = \frac{1}{x} - \log_e x \text{ is decreasing function}$$

$$f'(e) = \frac{1}{e} - 1 < 0$$

$$f'(e^2) = \frac{1}{e^2} - 2 < 0 \text{ is true}$$

$$(R) f''(x) = -\frac{1}{x} - \frac{1}{x^2} < 0 \text{ so, } f'(x) \text{ is decreasing}$$

So, (R) is false.

(S) is true.

SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : Let $f(x) = |x^2 - 1|$, $x \in [-2, 2] \Rightarrow f(-2) = f(2)$ and hence there must be at least one $c \in (-2, 2)$ so that $f'(c) = 0$.

and

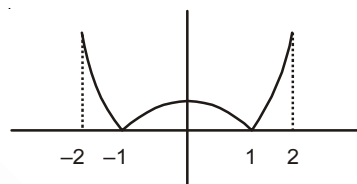
STATEMENT-2 : $f'(0) = 0$, where $f(x)$ is the function of S_1 .

Sol. Answer (4)

Statement-1 : $f(x)$ is not differentiable at $x = \pm 1$ hence

Rolle's theorem is not applicable consequently Statement-1 is false.

From graph $f(0)$, $a = 0$ i.e., Statement-2 is True.



2. STATEMENT-1 : Let $f(x)$ and $g(x)$ be two decreasing function then $f(g(x))$ must be an increasing function.

and

STATEMENT-2 : $f(g(2)) > f(g(1))$; where f and g are two decreasing function.

Sol. Answer (2)

Statement-1 : We have $f'(x) < 0$ and $g'(x) < 0$

Let $h(x) = f(g(x))$

$$\Rightarrow h'(x) = f'(g(x)) \cdot g'(x) \Rightarrow h'(x) > 0 \text{ hence } f(g(x)) \text{ is increasing}$$

$\Rightarrow$ Statement-1 is true.

Since $g(2) < g(1)$, as $g(x)$ is a decreasing function

$$\Rightarrow f[g(2)] > f[g(1)] \text{ as } f(x) \text{ is a decreasing function}$$

Hence Statement-2 is true but it is not correct explanation of statement-1.

3. STATEMENT-1 : $f(x) = x^3$ is a one-one function.

and

STATEMENT-2 : Any monotonic function is a one-one function.

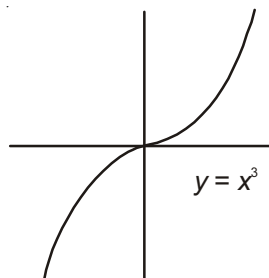
Sol. Answer (1)

Clearly from graph x^3 is one-one.

Statement-1 is true.

By monotonic function we mean either it is monotonic increasing or monotonic decreasing.

$\Rightarrow$ Any monotonic function is one-one hence Statement-2 is true and correct explanation of Statement-1.



4. STATEMENT-1 : $f(x) = |x - 1| + |x - 2| + |x - 3|$ has a local minima.

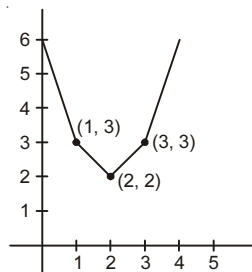
and

STATEMENT-2 : Any differentiable function $f(x)$ may have a local maxima or minima if $f'(x) = 0$ at some points.

Sol. Answer (2)

Statement-1 : $f(x) = |x - 1| + |x - 2| + |x - 3|$

$$f(x) = \begin{cases} -3x + 6, & x < 1 \\ -x + 4, & 1 \leq x < 2 \\ x, & 2 \leq x < 3 \\ 3x - 6, & x \geq 3 \end{cases}$$



$f(x)$ has minima at $x = 2$.

If $f'(x) = 0$ then it is either point of maxima or minima hence Statement-2 is true but not correct explanation.

5. STATEMENT-1 : Let a function $f(x)$ be satisfying $f'(x) = \frac{f(b) - f(a)}{b - a}$, $x \in (a, b) \Rightarrow f(x)$ is a differentiable function on $[a, b]$.

and

STATEMENT-2 : According to LMVT if $f(x)$ is differentiable on $[a, b]$ then $f'(x) = \frac{f(b) - f(a)}{(b - a)}$, $a < x < b$.

Sol. Answer (4)

Statement-1 : If $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) then

$$f'(x) = \frac{f(b) - f(a)}{b - a}, \forall x \in (a, b)$$

Hence Statement-1 is false.

Statement-2 is True.

6. STATEMENT-1 : Let $f(x) = e^x$ and $g(x) = e^{2x} \Rightarrow f(x) \cdot g(x)$ be an increasing function.

and

STATEMENT-2 : Multiplication of two increasing function is an increasing function.

Sol. Answer (3)

Statement-1 : $f(x) = e^x, g(x) = e^{2x}$

$$h(x) = f(x) \cdot g(x) = e^{3x}$$

$$h'(x) = 3e^{3x}$$

$\Rightarrow h'(x) > 0 \forall x \in R$ hence $h(x)$ is increasing i.e., Statement-1 is True.

It is not necessary that multiplication of two increasing function is increasing hence Statement-2 is false.

7. Consider the function $f(x) = 2x^3 + 3x^2 + 6x$.

STATEMENT-1 : $f'(x) > 0, \forall x \in R$

and

STATEMENT-2 : f is an increasing function.

Sol. Answer (1)

$$f(x) = 2x^3 + 3x^2 + 6x$$

$$f'(x) = 6x^2 + 6x + 6 = 6(x^2 + x + 1)$$

$$= 6 \left[\left(x + \frac{1}{2} \right)^2 + \frac{3}{4} \right]$$

$f'(x) > 0$ Statement-1 is true.

i.e., Statement-2 is true and correct explanation of statement-1.

SECTION - E

Matrix-Match Type Questions

1. Match the greatest value of the function in column I to column II.

Column-I

(A) $y = \sqrt{100 - x^2}$ on $[-6, 8]$

(B) $y = 2 \tan x - \tan^2 x$ on $\left[0, \frac{\pi}{2}\right)$

(C) $y = \tan^{-1} \frac{1-x}{1+x}$ on $[0, 1]$

(D) $y = \frac{a^2}{x} + \frac{b^2}{1-x}$ on $(0, 1)$, $a > 0, b > 0$

Column-II

(p) No greatest value

(q) 10

(r) $\frac{\pi}{4}$

(s) 1

Sol. Answer A(q), B(s), C(r), D(p)

(A) $y = \sqrt{100 - x^2}$, domain $x \in [-10, 10]$

$$\frac{dy}{dx} = \frac{-x}{\sqrt{100 - x^2}}$$

critical points of $y = f(x)$ are $x = 0, \pm 10$

We find that

$$f(0) = 10$$

$$f(10) = 0$$

$$f(-10) = 0$$

$$y_{\max} = 10$$

$$(B) \quad y = 2 \tan x - \tan^2 x$$

$$\frac{dy}{dx} = 2 \sec^2 x - 2 \tan x \cdot \sec^2 x$$

for maxima-minima of y

$$\frac{dy}{dx} = 0 \Rightarrow x = \frac{\pi}{4}$$

$$\left(\frac{d^2 y}{dx^2} \right)_{x=\frac{\pi}{4}} < 0$$

$$y_{\max} = 1$$

$$(C) \quad y = \tan^{-1} \frac{1-x}{1+x}, \quad 0 \leq x \leq 1$$

$$\Rightarrow 0 \leq \frac{1-x}{1+x} \leq 1$$

$$0 \leq \tan^{-1} \left(\frac{1-x}{1+x} \right) \leq \frac{\pi}{4}$$

$$y_{\max} = \frac{\pi}{4}$$

$$(D) \quad y = \frac{a^2}{x} + \frac{b^2}{1-x} \Rightarrow \frac{dy}{dx} = -\frac{a^2}{x^2} + \frac{b^2}{(1-x)^2}$$

for maxima-minima of y

$$\frac{dy}{dx} = 0 \Rightarrow x = \frac{a}{a+b}$$

$$\left(\frac{d^2 y}{dx^2} \right)_{x=\frac{a}{a+b}} > 0 \text{ i.e., it is minima.}$$

hence no maximum value.

2. Let the function defined in Column-I have domain $\left(0, \frac{\pi}{2}\right)$ then match the followings.

Column-I

(A) $x^2 + 2\cos x + 2$ has

(B) $9x - 4 \tan x$ has

(C) $\left(\frac{1}{2} - x\right)\cos x + \sin x - \frac{x^2 - x}{4}$ has

(D) $\left(\frac{1}{2} - x\right)\cos \pi(x+3) + \frac{1}{\pi}\sin \pi(x+3)$ has

Column-II

(p) Local maximum at $\cos^{-1}\left(\frac{2}{3}\right)$

(q) Maximum at $x = \frac{1}{2}$

(r) No local extremum

(s) Minima at $x = 1$

Sol. Answer A(r), B(p), C(q), D(q, s)

(A) $f(x) = x^2 + 2\cos x + 2 \Rightarrow f'(x) = 2x - 2\sin x = 2(x - \sin x)$

For maxima–minima of $f(x)$

$$f'(x) = 0 \Rightarrow x = \sin x \Rightarrow \text{no value of } x \text{ lies in } \left(0, \frac{\pi}{2}\right)$$

i.e., No extremum.

(B) $f(x) = 9x - 4 \tan x$

$$f'(x) = 9 - 4 \sec^2 x$$

For maxima–minima of $f(x)$

$$f'(x) = 0$$

$$\sec^2 x = \frac{9}{4} \Rightarrow \sec x = \frac{3}{2} \Rightarrow \cos x = \frac{2}{3}$$

$$x = \cos^{-1}\left(\frac{2}{3}\right)$$

$$f''(x) = -8 \sec^2 x \tan x$$

$$f''\left(\cos^{-1}\frac{2}{3}\right) < 0$$

Hence $f(x)$ attains maxima for $x = \cos^{-1}\left(\frac{2}{3}\right)$

$$(C) f(x) = \left(\frac{1}{2} - x\right) \cos x + \sin x - \frac{x^2 - x}{4}$$

$$\begin{aligned} \Rightarrow f'(x) &= \frac{-\sin x}{2} + x \sin x - \cos x + \cos x - \frac{2x-1}{4} \\ &= -\frac{\sin x}{2} + x \sin x - \frac{2x-1}{4} = \sin x \left(x - \frac{1}{2}\right) - \left(\frac{x}{2} - \frac{1}{4}\right) \end{aligned}$$

For maxima-minima of $f(x)$

$$f'(x) = 0 \Rightarrow x = \frac{1}{2}$$

$$\text{Now } f''(x) = \left(x - \frac{1}{2}\right) \cos x + \sin x - \frac{1}{2}$$

$$f''\left(\frac{1}{2}\right) = \sin \frac{1}{2} - \frac{1}{2} < 0$$

Hence $f(x)$ attains maximum value at $x = \frac{1}{2}$

$$(D) f(x) = \left(\frac{1}{2} - x\right) \cos \pi(x+3) + \frac{1}{\pi} \sin \pi(x+3)$$

$$f'(x) = \pi \left(x - \frac{1}{2}\right) \sin \pi(x+3)$$

For maxima-minima of $f(x)$

$$f'(x) = 0 \Rightarrow x = \frac{1}{2}, 1$$

$$f''(x) = \pi^2 \left(x - \frac{1}{2}\right) \cos \pi(x+3) + \pi \sin \pi(x+3)$$

$$f''\left(\frac{1}{2}\right) = -\pi < 0 \Rightarrow f(x) \text{ attains maximum value at } x = \frac{1}{2}$$

$$f''(1) = \frac{\pi^2}{2} > 0 \Rightarrow f(x) \text{ attains minima in the neighbourhood of } x = 1$$

3. Match the followings

Column I

- (A) The length of interval in which $f(x) = x^2 e^{-x}$ is monotonic increasing is
- (B) If $y = kx^3 + 3x^2 + (2k+1)x + 1000$ is strictly increasing for all values of x , then the least of positive integral value which k can attain, is

Column II

- (p) 1
- (q) 2

- (C) If the sum of the squares of intercepts on axes made by a tangent at any point on the curve $x^{2/3} + y^{2/3} = a^{2/3}$ is a^k then k is
- (D) If $\frac{\log x}{x}$ attains maximum value at $x = k$ then k is
- (r) Irrational number
- (s) Greater than 1
- (t) An integer

Sol. Answer A(q, s, t), B(p, t), C(q, s, t), D(r, s)

(A) $f(x) = x^2 e^{-x} \Rightarrow f'(x) = -x e^{-x}(x - 2) > 0$ for $x \in (0, 2)$

hence the length of interval is 2.

(B) $y = kx^3 + 3x^2 + (2k + 1)x + 1000$

$\Rightarrow y' = f'(x) = 3kx^2 + 6x + (2k + 1) > 0, \forall x \in R$

$\Rightarrow k > 0$ and $6^2 - 12k(2k + 1) < 0$

$\Rightarrow k > 1$ we observe that for $k = 1, f(x) = (x + 1)^3 + 999$, which is clearly monotonic increasing.

Hence the least of positive integer which k attain is 1.

(C) Tangent at point (α, β) is

$$\frac{x}{\alpha^{1/3}} + \frac{y}{\beta^{1/3}} = a^{2/3}$$

Now, $a^3 \left(\alpha^{\frac{2}{3}} + \beta^{\frac{2}{3}} \right) = a^k$

$\Rightarrow a^{\frac{4}{3}} \cdot a^{\frac{2}{3}} = a^k$

$\Rightarrow a^2 = a^k \Rightarrow k = 2$

(D) $y = \frac{\log x}{x} \Rightarrow y' = -\frac{(\log x - 1)}{x^2}$

The point of maximum is e

Which is an irrational number greater than 1.

4. Match the followings according to the given conditions.

Column I

(A) Let k be the number which cannot be the slope of a tangent to the curve $x^2 + y^2 - 2x + 2y + 2xy = 1$, then k is

(B) $f(x) = 2\sec x - \sqrt{3} \tan x$, then minimum value of $f(x)$ is

(C) If the function $f(x) = \frac{ax + b}{(x - 1)(x - 4)}$ has a local minima at

$(2, -1)$, then a is

(D) Let $f(x) = \max\{|\sin x|, |\cos x|\} \forall x \in R$ then minimum value of $f(x)$ is

Column II

(p) Less than 1

(q) 1

(r) Whole number

(s) Irrational number

(t) $\frac{1}{\sqrt{2}}$

Sol. Answer A(q, r), B(q, r), C(q, r), D(p, s, t)

$$(A) \quad x^2 + y^2 - 2x - 2y - 2xy = 1 \quad \dots(1)$$

$$\Rightarrow (x - y)^2 = 2x + 2y + 1$$

This is a parabolic curve with slope of axis = 1, therefore there can be no tangent with slope = 1

Differentiating (1) w.r.t. x

$$\frac{dy}{dx} = \frac{x - y - 1}{x - y + 1}$$

$$\text{As } \frac{dy}{dx} = 1, \text{ we have } x - y + 1 = x - y - 1$$

$$\Rightarrow -1 = 1 \text{ (not possible)}$$

$$(B) \text{ The minimum value of } f(x) = a \sec x - b \tan x \text{ is } \sqrt{a^2 - b^2}$$

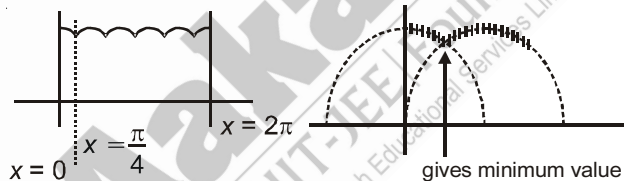
$$\text{Hence minimum of } 2 \sec x - \sqrt{3} \tan x \text{ is } \sqrt{2^2 - (\sqrt{3})^2} = 1$$

$$(C) \quad f(2) = -1 \quad \Rightarrow \quad 2a + b = 2 \quad \dots(1)$$

$$f(2) = 0 \quad \Rightarrow \quad b = 0 \quad \dots(2)$$

$$\text{From (1) and (2) } 2a = 2 \quad \Rightarrow \quad a = 1$$

$$(D) \quad y = \text{maximum } \{|\sin x|, |\cos x|\}$$



$$f_{\min} = f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

5. Match the interval of monotonicity of functions in column I to column II.

Column-I

$$(A) \quad y = x - e^x$$

$$(B) \quad y = \log(x + \sqrt{x^2 + 1})$$

$$(C) \quad y = x\sqrt{4x - x^2}$$

$$(D) \quad y = \frac{10}{4x^3 - 9x^2 + 6x}$$

Column-II

$$(p) \quad (-\infty, \infty)$$

$$(q) \quad (0, 3)$$

$$(r) \quad (0, \infty)$$

$$(s) \quad (1, \infty)$$

Sol. Answer A(q, r, s), B(p, q, r, s), C(q), D(s)

$$(A) \quad y = x - e^x$$

$$\frac{dy}{dx} = 1 - e^x$$

$$\begin{cases} \frac{dy}{dx} < 0, & \text{if } x > 0 \\ \frac{dy}{dx} > 0, & \text{if } x < 0 \end{cases}$$

i.e., y is monotonic in $(0, 3)$, $(0, \infty)$ and $(1, \infty)$

(B) $y = \log(x + \sqrt{x^2 + 1})$

$$\frac{dy}{dx} = \left(\frac{1}{x + \sqrt{x^2 + 1}} \right) \left(1 + \frac{2x}{2\sqrt{x^2 + 1}} \right) = \frac{1}{\sqrt{x^2 + 1}}$$

$$\Rightarrow \frac{dy}{dx} > 0 \quad \forall x \in R$$

(C) $y = x\sqrt{4x - x^2}$

$$\Rightarrow \frac{dy}{dx} = \sqrt{4x - x^2} + x \cdot \frac{1}{2\sqrt{4x - x^2}} \cdot (4 - 2x) = \frac{4x - x^2 + 2x^2 - x^2}{\sqrt{4x - x^2}} = \frac{2(3x - x^2)}{\sqrt{4x - x^2}}$$

$$\Rightarrow \frac{dy}{dx} > 0, \quad \forall x \in (0, 3)$$

(D) $y = \frac{10}{4x^3 - 9x^2 + 6x}$

$$\Rightarrow \frac{dy}{dx} = -\frac{10(12x^2 - 18x + 6)}{(4x^3 - 9x^2 + 6x)^2} \Rightarrow \frac{dy}{dx} = -\frac{60(2x^2 - 3x + 1)}{(4x^3 - 9x^2 + 6x)^2}$$

$$\Rightarrow \frac{dy}{dx} = -60 \frac{(x-1)(2x-1)}{(4x^3 - 9x^2 + 6x)^2} \Rightarrow \frac{dy}{dx} = \begin{cases} < 0, & \left(-\infty, \frac{1}{2}\right) \cup (1, \infty) \\ > 0, & \left(\frac{1}{2}, 1\right) \end{cases}$$

6. Match the extrema of the functions in column I to column II.

Column-I

(A) $y = x + \sqrt{1 - x^2}$

(B) $y = x^2 e^{-x}$

(C) $y = x\sqrt{2 - x^2}$

(D) $y = \frac{e^{ax} + e^{-ax}}{2}$

Column-II

(p) $y_{\min} = 0$

(q) $y_{\min} = 1$

(r) $y_{\max} = \sqrt{2}$

(s) $y_{\min} = -1$

Sol. Answer A(r, s), B(p), C(s), D(q)

(A) $f(x) = x + \sqrt{1-x^2}$

$$x = \cos\theta$$

$$f(x) = \cos\theta + |\sin\theta|, \sin\theta \geq 0$$

$$f(x)_{\max} = \sqrt{2} \quad \theta \in [0, \pi]$$

Now,

$$f(x) = \cos\theta - \sin\theta \quad \pi < \theta < 2\pi$$

$$f(x)_{\min} = -1$$

(B) $y = x^2 e^{-x}$

$$\Rightarrow \frac{dy}{dx} = x e^{-x} (2 - x)$$

at $x = 0$ sign of $\frac{dy}{dx}$ changes from negative to positive

Hence $y = f(x)$ attains minima at $x = 0$ and $\min f(x) = 0$

at $x = 2$ sign of $\frac{dy}{dx}$ changes from positive to negative

Hence $y = f(x)$ attains maxima at $x = 2$ and $y_{\max} = \frac{4}{e^2}$

$$\therefore y_{\min} = 0, y_{\max} = \frac{4}{e^2}$$

(C) $y = x\sqrt{2-x^2}$, domain $x \in [-\sqrt{2}, \sqrt{2}]$

$$\Rightarrow \frac{dy}{dx} = \frac{-x^2}{\sqrt{2-x^2}} + \sqrt{2-x^2} \Rightarrow \frac{dy}{dx} = \frac{2(1-x^2)}{\sqrt{2-x^2}}$$

$$\Rightarrow \frac{dy}{dx} = \begin{cases} > 0, & -1 < x < 1 \\ < 0, & x \in (-\sqrt{2}, -1) \cup (1, \sqrt{2}) \end{cases}$$

In the neighbourhood of $x = -1$, $\frac{dy}{dx}$ changes from negative to positive hence y attains minima at $x = -1$.

In the neighbourhood of $x = 1$, $\frac{dy}{dx}$ changes from positive to negative hence y attains maxima at $x = 1$.

$$y_{\min} = -1, y_{\max} = 1$$

$$(D) y = \frac{e^{ax} + e^{-ax}}{2}$$

From A.M. – G. M. inequality, we have

$$\frac{e^{ax} + e^{-ax}}{2} \geq 1$$

$$y_{\min} = 1.$$

7. Match the value of the function at local maxima in column I to column II.

Column-I

$$(A) x^2 e^{-x}$$

$$(B) \frac{4x}{x^2 + 4}$$

$$(C) -x^2 (x-2)^{2/5}$$

$$(D) \frac{14}{x^4 - 8x^2 + 2}$$

Column-II

$$(p) -1$$

$$(q) 0$$

$$(r) f(2)$$

$$(s) 1$$

Sol. Answer A(r), B(r, s), C(q, r), D(p, r)

$$(A) y = x^2 e^{-x}$$



$$\frac{dy}{dx} = -x^2 e^{-x} + 2x e^{-x} = x(2-x)e^{-x} = -e^{-x}(x^2 - 2x)$$

We observe that $f'(x)$ changes its sign from +ve to -ve in the neighbourhood of $x = 2$ and from -ve to +ve in the neighbourhood of $x = 0$

$\therefore f(2)$ is maxima.

$$(B) y = \frac{4x}{x^2 + 4} \Rightarrow y = \frac{4}{x + \frac{4}{x}}$$

A.M. – G.M. inequality

$$x + \frac{4}{x} \geq 4, \forall x > 0$$

$$\Rightarrow \frac{4}{x + \frac{4}{x}} \leq 1$$

$$y_{\max} = 1$$

$$(C) y = -x^2 (x-2)^{2/5}$$

$$\frac{dy}{dx} = -\frac{2}{5} \frac{x^2}{(x-2)^{3/5}} - 2x(x-2)^{2/5} = -2x \left[\frac{x+5x-10}{5(x-2)^{3/5}} \right] = -\frac{2x(6x-10)}{5(x-2)^{3/5}}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{4x(3x-5)}{5(x-2)^{3/5}}$$

Critical points of the given function are $0, \frac{5}{3}, 2$

$$f(0) = 0$$

$$f(2) = 0$$

$$f\left(\frac{5}{3}\right) = -\frac{25}{3}\left(\frac{1}{9}\right)^{1/5}$$

$$y_{\max} = 0.$$

$$(D) \quad y = \frac{14}{x^4 - 8x^2 + 2}$$

y is maximum when $x^4 - 8x^2 + 2$ is minimum

$$\text{Let } z = x^4 - 8x^2 + 2 \Rightarrow \frac{dz}{dx} = 4x^3 - 16x$$

For maxima-minima of z

$$\frac{dz}{dx} = 0 \Rightarrow x = 0, \pm 2$$

$$\left(\frac{d^2z}{dx^2}\right) = 12x^2 - 16$$

$$\left(\frac{d^2z}{dx^2}\right)_{x=\pm 2} > 0 \Rightarrow z \text{ is minimum for } x = \pm 2$$

$$\text{Hence } y_{\max} = \frac{14}{16 - 32 + 2} = -1$$

SECTION - F

Integer Answer Type Questions

1. The ratio of absolute maxima and minima of $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$, ($x \in R$) is _____.

Sol. Answer (9)

The absolute maxima = 3 and the absolute minima = $\frac{1}{3}$

The required ratio = $\frac{3}{1/3} = 9$

[we know $\frac{1}{3} \leq \frac{x^2 - x + 1}{x^2 + x + 1} \leq 3$]

2. Consider the function $f(x) = x + \frac{1}{x}$, $x \in \left(\frac{1}{2}, \frac{9}{2}\right)$. If α is the length of interval of decrease and β be the length of interval of increase, then $\frac{\beta}{\alpha}$ is _____.

Sol. Answer (7)

$f(x) = x + \frac{1}{x}$, $x \in \left(\frac{1}{2}, \frac{9}{2}\right)$ is decreasing in $\left(\frac{1}{2}, 1\right)$ and increasing in $\left(1, \frac{9}{2}\right)$,

$$\alpha = 1 - \frac{1}{2} = \frac{1}{2}, \quad \beta = \frac{9}{2} - 1 = \frac{7}{2}$$

$$\therefore \frac{\beta}{\alpha} = 7$$

3. If the function $f(x) = \frac{1}{1+x^2}$ is downward concave in (α, β) then $[\beta - \alpha]$ is _____ ([.] is greatest integer function.)

Sol. Answer (1)

$$f''(x) = \frac{6\left(x^2 - \frac{1}{3}\right)}{(1+x^2)^3} < 0 \text{ in } \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right). \text{ Hence } \alpha = \frac{1}{\sqrt{3}}, \beta = \frac{1}{\sqrt{3}}$$

$$[\beta - \alpha] = \left[\frac{1}{\sqrt{3}} - \left(-\frac{1}{\sqrt{3}}\right)\right] = \left[\frac{2}{\sqrt{3}}\right] = 1$$

4. The slope of the tangent to the curve $(y - x^5)^2 = x(1 + x^2)^2$ at the point (1, 3) is

[JEE(Advanced)-2014]

Sol. Answer (8)

$$(y - x^5)^2 = x(1 + x^2)^2$$

$$2(y - x^5)\left(\frac{dy}{dx} - 5x^4\right) = (1 + x^2)^2 + 2x(1 + x^2) \cdot 2x$$

$$x = 1, y = 3$$

$$2(3 - 1)\left(\frac{dy}{dx} - 5\right) = (1 + 1)^2 + 4(1 + 1)$$

$$4\left(\frac{dy}{dx} - 5\right) = 12 \Rightarrow \frac{dy}{dx} - 5 = 3 \Rightarrow \frac{dy}{dx} = 8$$

5. A cylindrical container is to be made from certain solid material with the following constraints: It has a fixed inner volume of $V \text{ mm}^3$, has a 2 mm thick solid wall and is open at the top. The bottom of the container is a solid circular disc of thickness 2 mm and is of radius equal to the outer radius of the container.

If the volume of the material used to make the container is minimum when the inner radius of the container

is 10 mm, then the value of $\frac{V}{250\pi}$ is

[JEE(Advanced)-2015]

Sol. Answer (4)

Volume of material used

$$V_1 = \pi(r+2)^2h - \pi r^2h + \pi(r+2)^2 \cdot 2$$

$$= 4\pi h(r+1) + 2\pi(r+2)^2$$

$$= \frac{4V}{r^2}(r+1) + 2\pi(r+2)^2 \quad \left(\because h = \frac{V}{\pi r^2} \right)$$

$$\frac{dV_1}{dr} = 4V \left(-\frac{1}{r^2} - \frac{2}{r^3} \right) + 4\pi(r+2)$$

$$\text{when } r = 10, \quad \frac{dV_1}{dr} = 0$$

$$\Rightarrow 4V \left(\frac{-1}{100} - \frac{-2}{1000} \right) + 48\pi = 0$$

$$\Rightarrow 4V \left(\frac{12}{1000} \right) = 48\pi$$

$$\therefore \frac{V}{250\pi} = 4$$

SECTION - G**Multiple True-False Type Questions**

1. STATEMENT-1 : If two sides of a triangle are given, then its area will be maximum if the angle between the given sides be $\frac{\pi}{2}$.

STATEMENT-2 : The function $f(x) = \frac{1}{1+x^2}$ increases in the interval $(0, \infty)$.

STATEMENT-3 : The length of the subnormal of the curve $y^3 = 12ax$ at $y = 1$ is $4a$.

(1) T F T

(2) T T T

(3) F F F

(4) F F T

Sol. Answer (1)

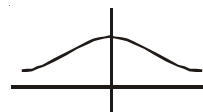
STATEMENT-1 True, Area = $\frac{1}{2}ab \sin C \leq \frac{1}{2}ab \sin C$ [$\sin C \leq 1$]

$$\text{Area maximum} = \frac{1}{2}ab \left[\sin C = 1 \text{ at } C = \frac{\pi}{2} \right]$$

STATEMENT-2 False, $y = f(x) = \frac{1}{1+x^2}$

$f(x)$ decreases in $(0, \infty)$

STATEMENT-3 True, subnormal = $yy' = y \cdot \left(\frac{4a}{y^2} \right) = 4a$, at $y = 1$



2. STATEMENT-1 : The maximum value of $3\sin x + 4\cos x$ is 7.

STATEMENT-2 : In the interval $[-1, 1]$ the greatest value of $f(x) = \frac{x}{4+x+x^2}$ is $\frac{1}{5}$.

STATEMENT-3 : The greatest value of $4\sin^2 x + 3\cos^2 x$ is 4.

(1) T F T

(2) T T T

(3) F F F

(4) F F T

Sol. Answer (4)

STATEMENT-1 False, Maximum value of $3\sin x + 4\cos x = \sqrt{3^2 + 4^2} = 5$

STATEMENT-2 False, $f(x) = \frac{x}{4+x+x^2}$, $f'(x) = -\frac{x^2-4}{(x^2+x+4)^2} > 0$ for $x \in (-2, 2)$

$\Rightarrow f(x)$ is increasing in $[-1, 1]$

$$f_{\max} = f(1) = \frac{1}{6}$$

STATEMENT-3 True, $f(x) = 4\sin^2 x + 3\cos^2 x = 3 + \sin^2 x \leq 4$.

Hence $f(x)_{\max} = 4$

SECTION - H

Aakash Challengers Questions

1. The least natural number a for which $x + \frac{a}{x^2} > 2 \forall x \in (0, \infty)$ is _____

Sol. $x + \frac{a}{x^2} > 2$

$$x^3 - 2x^2 + a > 0$$

$$\text{Let } f(x) = x^3 - 2x^2 + a$$

$$\Rightarrow f'(x) = 3x^2 - 4x \quad \Rightarrow f'(x) = x(3x - 4)$$

For maximum or minimum value of $f(x) = 0 \Rightarrow x = \frac{4}{3}, 0$

$$f''(x) = 6x - 4$$

$$\Rightarrow f''\left(\frac{4}{3}\right) = 4 > 0 \quad \Rightarrow f(x) \text{ attains minimum value at } x = \frac{4}{3}$$

$$\text{and } (f(x))_{\min} = f\left(\frac{4}{3}\right) = -\frac{32}{27} + a$$

for $f(x) > 0$

$$\Rightarrow f\left(\frac{4}{3}\right) > 0 \quad \Rightarrow a > \frac{32}{27}, \text{ i.e., least natural number value of } a = 2.$$

2. Any tangent at a point $P(x, y)$ to the ellipse $\frac{x^2}{8} + \frac{y^2}{18} = 1$ meets the positive co-ordinate axes in the points A and B such that the area of the ΔOAB is least, then the point P is of the form (m, n) where $m + n + 10$ is

Sol. Tangent at P is $\frac{x \cdot 2\sqrt{2} \cos \theta}{8} + \frac{y \cdot 3\sqrt{2} \sin \theta}{18} = 1$

$$\Rightarrow \frac{x}{2\sqrt{2} \sec \theta} = \frac{y}{3\sqrt{2} \sec \theta} = 1$$

$$\therefore A : (2\sqrt{2} \sec \theta, 0), B : (0, 3\sqrt{2} \csc \theta)$$

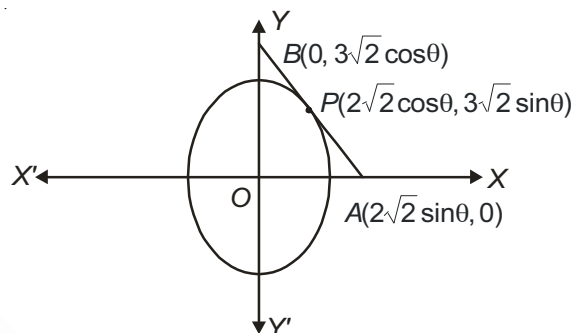
$$\text{ar}(\Delta OAB) = \frac{1}{2} \times 2\sqrt{2} \sec \theta \times 3\sqrt{2} \csc \theta = \frac{12}{\sin 2\theta}$$

$$\Rightarrow \text{ar}(\Delta OAB) \text{ is least on } \sin 2\theta = 1$$

$$\Rightarrow 2\theta = \frac{\pi}{2} \quad \Rightarrow \theta = \frac{\pi}{4}$$

$$\therefore P = \left(2\sqrt{2} \cdot \frac{1}{\sqrt{2}}, 3\sqrt{2} \times \frac{1}{\sqrt{2}} \right) = (2, 3) = (m, n)$$

$$\Rightarrow m + n + 10 = 2 + 3 + 10 = 15$$



3. If the tangent at $P(1, 1)$ on the curve $y^2 = x(2-x)^2$ meets the curve again at Q , then the points θ is of the form $\left(\frac{3a}{b}, \frac{a}{2b} \right)$, where $a^2 + b^2$ is

Sol. Given equation is $y^2 = x(2-x)^2$... (i)

$$\begin{aligned} \therefore 2y \frac{dy}{dx} &= (2-x)^2 - 2x(2-x) \\ &= 4 - 4x + x^2 - 4x + 2x^2 \\ &= 3x^2 - 8x + 4 \end{aligned}$$

$$\Rightarrow \frac{dy}{dx} = \frac{3x^2 - 8x + 4}{2y}$$

$$\therefore m = \left. \frac{dy}{dx} \right|_{(1,1)} = \frac{3-8+4}{2 \cdot 1} = -\frac{1}{2}$$

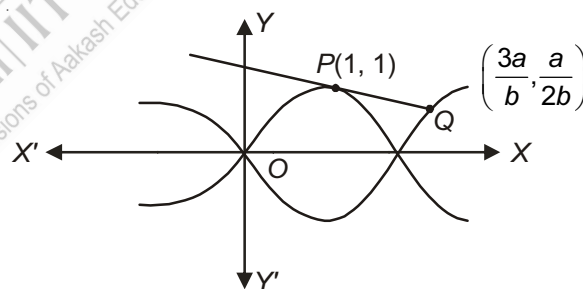
$$\text{Tangent at } P \text{ is } y - 1 = -\frac{1}{2}(x - 1)$$

$$\Rightarrow 2y - 2 = -x + 1$$

$$\Rightarrow x + 2y = 3$$

From (1) & (2), we get

$$\left(\frac{3-x}{2} \right)^2 = x(x-2)^2$$



$$\Rightarrow (3-x)^2 = 4x(x-2)^2$$

$$\Rightarrow 4x^3 - 17x^2 + 22x - 9 = 0$$

$$\Rightarrow (x-1)(4x^2 - 13x + 9) = 0$$

$$\Rightarrow (x-1)(x-1)(4x-9) = 0$$

$$\Rightarrow x = 1, \frac{9}{4}$$

$$\Rightarrow y = \frac{3-x}{2} = \frac{3-\frac{9}{4}}{2} = \frac{3}{8}$$

$$\Rightarrow Q \equiv \left(\frac{9}{4}, \frac{3}{8}\right) \equiv \left(\frac{3a}{b}, \frac{a}{2b}\right)$$

$$\Rightarrow 3a = 9, b = 4$$

$$\Rightarrow a = 3, b = 4$$

$$\therefore a^2 + b^2 = 3^2 + 4^2 = 25$$

4. The curve $y = ax^3 + bx^2 + cx$ is inclined at 45° to x-axis at $(0, 0)$ but it touches x-axis at $(1, 0)$, then $a + b + c + 10$ is

Sol. Given curve is $y = ax^3 + bx^2 + cx$

$$\Rightarrow \frac{dy}{dx} = 3ax^2 + 2bx + c$$

$$\text{Given } \tan 45^\circ = 1$$

$$\Rightarrow 3ax^2 + 2bx + c = 1$$

$$\text{Now, } \left(\frac{dy}{dx}\right)_{(0,0)} \Rightarrow c = 1$$

$$\left(\frac{dy}{dx}\right)_{(1,0)} \Rightarrow 3a + 2b + c = 0 \Rightarrow 3a + 2b + 1 = 0$$

$$\text{Also, } a + b + c = 0$$

$$\therefore 3a + 2b + 1 = 0$$

$$a + b + 1 = 0$$

$$\Rightarrow \underline{2a + 2b + 2 = 0}$$

$$\underline{a - 1 = 0}$$

$$\Rightarrow a = 1 \therefore b = -2$$

$$\Rightarrow a + b + c + 10 = 1 - 2 + 1 + 10 = 10$$

5. Let $f(x) = 2x^3 + ax^2 + bx - 3\cos^2 x$ is an increasing function for all $x \in R$ such that $ma^2 + nb + 18 < 0$ then the value of $m + n + 7$ is

Sol. Given $f(x) = 2x^3 + ax^2 + bx - 3\cos^2 x$

$$\Rightarrow f'(x) = 6x^2 + 2ax + b + 3\sin 2x > 0$$

$$\Rightarrow 6x^2 + 2ax + b - 3 > 0 \quad (\because \sin 2x \geq -1)$$

$$\Rightarrow 6x^2 + 2ax + (b - 3) > 0$$

$$D < 0$$

$$\Rightarrow 4a^2 - 24(b - 3) < 0$$

$$\Rightarrow a^2 - 6b + 18 < 0$$

$$\Rightarrow m = 1, n = -6$$

$$\therefore m + n + 7 = 1 - 6 + 7 = 8 - 6 = 2$$

6. Let $g(x) = 2f\left(\frac{x}{2}\right) + f(2-x)$ and $f''(x) < 0 \forall x \in (0, 2)$. If $g(x)$ increases in (a, b) and decreases in (c, d) , then the value of $a + b + c + d - \frac{2}{3}$ is

Sol. We have $g'(x) = 2 \times \frac{1}{2} f'\left(\frac{x}{2}\right) - f'(2-x) = f'\left(\frac{x}{2}\right) - f'(2-x)$

$$\Rightarrow f'(x) \text{ is increase in } (0, 2) (\because f''(x) < 0 \forall x \in (0, 2))$$

$$\text{Let } \frac{x}{2} > 2-x$$

$$\frac{3x}{2} > 2$$

$$\Rightarrow f'\left(\frac{x}{2}\right) < f'(2-x), x > \frac{4}{3}$$

$$\Rightarrow x > \frac{4}{3}$$

$$\Rightarrow g'(x) < 0$$

$$\Rightarrow g(x) \text{ decrease in } \left(\frac{4}{3}, 2\right)$$

$$\text{Also, } \frac{x}{2} < 2-x$$

$$\Rightarrow f'\left(\frac{x}{2}\right) > f'(2-x), x < \frac{4}{3}$$

$$\Rightarrow g'(x) > 0$$

$$\Rightarrow g(x) \text{ increase in } \left(0, \frac{4}{3}\right)$$

$$\text{Thus } (a, b) \equiv \left(0, \frac{4}{3}\right)$$

$$\text{And } (c, d) \equiv \left(\frac{4}{3}, 2\right)$$

$$\Rightarrow a + b + c + d - \frac{2}{3} = 0 + \frac{4}{3} + \frac{4}{3} + 2 - \frac{2}{3} = \frac{8}{3} - \frac{2}{3} + 2 = 2 + 2 = 4$$

7. If the equation $3x^2 + 4ax + b = 0$ has at least one root in $(0, 1)$ such that $La + Mb + N = 0$, where L, M, N are co-prime numbers, then the value of $L + M + N + LMN$ is

Sol. Let $f(x) = x^3 + 2ax^2 + bx$

Clearly $f(x)$ is continuous in $[0, 1]$ and differentiable in $(0, 1)$

$$\text{Also, } f(0) = 0 = f(1)$$

Thus, all the conditions of Rolle's theorem are satisfied

$$\therefore f(1) = 0 \Rightarrow 1 + 2a + b = 0$$

$$\Rightarrow L = 1, M = 2, N = 1$$

$$\therefore L + M + N + LMN = 1 + 2 + 1 + 2 = 6$$

8. The integral value of 'b' for which the function $f(x) = (b^2 - 3b + 2)(\cos^2 x - \sin^2 x) + (b - 1)x + \sin(b^2 + b + 1)$ does not possess any stationary point is

Sol. Given $f(x) = (b^2 - 3b + 2)\cos 2x + (b - 1)x + \sin(b^2 + b + 1)$

$$\Rightarrow f'(x) = -2(b - 1)(b - 2)\sin 2x + (b - 1) \neq 0 \text{ for any } x \in R$$

$$\Rightarrow (b - 1)(1 - (b - 2)(2\sin 2x)) \neq 0$$

$$\Rightarrow b \neq 1 \& \left| \frac{1}{2(b - 2)} \right| > 1$$

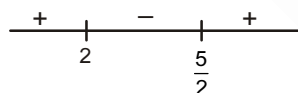
$$\Rightarrow \frac{1}{2(b - 2)} > 1, \frac{1}{2(b - 2)} < -1$$

$$\Rightarrow \frac{1}{b - 2} - 2 > 0$$

$$\Rightarrow \frac{1 - 2b + 4}{b - 2} > 0$$

$$\Rightarrow \frac{5 - 2b}{b - 2} > 0$$

$$\Rightarrow \frac{2b - 5}{b - 2} < 0$$



$$\Rightarrow b \in \left(2, \frac{5}{2}\right)$$

$$\Rightarrow b \in \left(\frac{3}{2}, 2\right) \cup \left(2, \frac{5}{2}\right)$$

$$\Rightarrow \frac{1}{b - 2} < -2$$

$$\Rightarrow \frac{1}{b - 2} + 2 < 0$$

$$\Rightarrow \frac{1 + 2b - 4}{b - 2} < 0$$

$$\Rightarrow \frac{2b - 3}{b - 2} < 0$$



$$\Rightarrow b \in \left(\frac{3}{2}, 2\right)$$

When $b = 2$, $f(x) = x + \sin(7)$

$$\Rightarrow f'(x) = 1 \neq 0$$

$$\text{Thus, } b \in \left(\frac{3}{2}, \frac{5}{2}\right)$$

$$\Rightarrow b = 2$$

9. If the point on $y = x \tan \alpha - \frac{ax^2}{32 \cos^2 \alpha}$ ($\alpha > 0$), where the tangent is parallel to $y = x$ has an ordinate $\frac{4}{a}$ then $4 \sin^2 \alpha$ equals _____.

Sol. We have,

$$y = x \tan \alpha - \frac{ax^2}{32 \cos^2 \alpha} \Rightarrow \frac{dy}{dx} = \tan \alpha - \frac{ax}{16 \cos^2 \alpha}$$

tangent is parallel to $y = x$

$$\text{hence, } \tan \alpha - \frac{ax}{16 \cos^2 \alpha} = 1$$

$$\Rightarrow x = \frac{(\tan \alpha - 1)16 \cos^2 \alpha}{a} = \frac{16}{a}(\sin \alpha - \cos \alpha) \cos \alpha$$

The corresponding ordinate

$$y = \frac{16}{a}(\sin \alpha - \cos \alpha) \sin \alpha - \frac{a}{32 \cos^2 \alpha} \frac{256}{a^2} (\sin \alpha - \cos \alpha)^2 \cdot \cos^2 \alpha$$

$$\Rightarrow \frac{4}{a} = \frac{16}{a} \left[\sin^2 \alpha - \sin \alpha \cos \alpha - \frac{1}{2}(1 - 2 \sin \alpha \cos \alpha) \right]$$

$$\Rightarrow \frac{1}{4} = \left(\sin^2 \alpha - \frac{1}{2} \right) \Rightarrow \sin^2 \alpha = \frac{3}{4} \Rightarrow 4 \sin^2 \alpha = 3$$

10. The least value of 'a' for which the equation $\frac{4}{\sin x} + \frac{1}{1 - \sin x} = a$ has at least one solution in $\left(0, \frac{\pi}{2}\right)$ is _____.

Sol. Let $y = \frac{4}{\sin x} + \frac{1}{1 - \sin x}$; $x \in \left(0, \frac{\pi}{2}\right) \Rightarrow \frac{dy}{dx} = \cos x \left(-\frac{4}{\sin^2 x} + \frac{1}{(1 - \sin x)^2} \right)$

$$\frac{dy}{dx} = 0 \Rightarrow \cos x = 0 \text{ or } \frac{4}{\sin^2 x} = \frac{1}{(1 - \sin x)^2}$$

$$\Rightarrow x = \frac{\pi}{2} \text{ or } 4(1 - \sin x)^2 = \sin^2 x$$

$$\text{Since } x \in \left(0, \frac{\pi}{2}\right) \Rightarrow 4(1 - \sin x)^2 = \sin^2 x \Rightarrow \sin x = \frac{2}{3}$$

$$\left(\frac{d^2 y}{dx^2} \right)_{x = \sin^{-1} \left(\frac{2}{3} \right)} = 35 > 0$$

Hence the given a has minimum at $x = \sin^{-1} \left(\frac{2}{3} \right)$ in $\left(0, \frac{\pi}{2}\right)$ and minimum $a = 9$.

11. The curve $y = ax^3 + bx^2 + cx + 5$ touches the x -axis at $P(-2, 0)$ and cuts the y -axis at Q , where its gradient is 3. Then the value of $a + b + c + abc$ is

Sol. P lies on curve

$$\therefore -8a + 4b - 2c + 5 = 0 \quad \dots(i)$$

$$\text{Also, } \left(\frac{dy}{dx} \right)_{(-2, 0)} = 0$$

$$\Rightarrow (3ax^2 + 2bx + c)_{(-2, 0)} = 0$$

$$\Rightarrow 12a - 4b + c = 0 \quad \dots(ii)$$

$$\text{Also, } \left(\frac{dy}{dx}\right)_{(0, 5)} = 3$$

$$\Rightarrow c = 3$$

$$\text{Now, } -8a + 4b = 1$$

$$12a - 4b = -3$$

$$\Rightarrow a = \frac{-1}{2}, b = \frac{-3}{4}$$

12. Let $f(x) = \int_0^x \log_2(\log_3(\log_4(\cos t + a))) dt$ be increasing for all real value of x , then the range of 'a'

Sol. $\therefore$ Given $f(x) = \int_0^x \log_2(\log_3(\log_4(\cos t + a))) dt$

$$\Rightarrow f'(x) = \log_2(\log_3(\log_4(\cos x + a)))$$

Since $f(x)$ is a function, so $f'(x) \geq 0$

$$\Rightarrow \log_2(\log_3(\log_4(\cos x + a))) \geq 0 \quad \forall x \in R$$

Now, $f(x)$ is increasing for all x , provided it is defined

$$\Rightarrow \log_3(\log_4(\cos x + a)) > 0$$

$$\Rightarrow \log_4(\cos x + a) > 1$$

$$\Rightarrow \cos x + a > 4$$

$$\Rightarrow a > 4 - \cos x \quad \forall x \in R$$

$$\Rightarrow a > 5 \quad (\because -1 \leq \cos x \leq 1)$$

$$\Rightarrow a \in (5, \infty)$$

13. If $f''(x) > 0$ and $f'(1) = 0$ such that $g(x) = f(\cot^2 x + 2\cot x + 2)$ where $0 < x < \pi$, then $g(x)$ decreasing in (a, b) ,

where $a + b + \frac{\pi}{4}$ is

Sol. Given $g(x) = f(\cot^2 x + 2\cot x + 2)$

$$\Rightarrow g'(x) = f'(\cot^2 x + 2\cot x + 2) \{-2\cot x \operatorname{cosec}^2 x - 2\operatorname{cosec}^2 x\} < 0$$

$$\Rightarrow f'\{(\cot x + 1)^2 + 1\} (\cot x + 1) > 0 \quad \dots(A)$$

Given $f''(x) > 0$

$$\Rightarrow f'(x) \text{ is increase}$$

$$\Rightarrow f'(\cot x + 1)^2 + 1 > f'(1). (\because f'(1) = 0)$$

$$\Rightarrow g'(x) < 0 \text{ if } (\cot x + 1) > 0 \quad \text{from (A)}$$

$$\Rightarrow x \in \left(0, \frac{3\pi}{4}\right)$$

$$\Rightarrow (a, b) \equiv \left(0, \frac{3\pi}{4}\right)$$

$$\Rightarrow a + b + \frac{\pi}{4} = 0 + \frac{3\pi}{4} + \frac{\pi}{4} = \pi$$

14. If the equation $x^3 - 3x + k = 0$ has two distinct roots in $(0, 1)$, then the value of k is

Sol. Let α and β such that $\alpha \neq \beta$ be the roots of $f(x) = x^3 - 3x + k = 0$

Then $0 < \alpha < \beta < 1$ and $f(\alpha) = 0 = f(\beta)$

Also $f(x)$ is differentiable in $(0, 1)$ and continuous in $[0, 1]$ and $f(\alpha) = 0 = f(\beta)$

Thus, all the conditions of Rolle's theorem are satisfied.

Now, we have to show that $\exists$ a point $c \in (\alpha, \beta)$ such that $f'(c) = 0$

$$\Rightarrow 3c^2 - 3c = 0$$

$$\Rightarrow c = 0, 1. \notin (0, 1)$$

$\therefore$ Thus $k = 0$

15. Let f' be a real function whose derivative upto third order exist and for same pair $a, b \in R$ such that $a < b$,

$$\log \left(\frac{f(a) + f'(a) + f''(a)}{f(b) + f'(b) + f''(b)} \right) = a - b \text{ then there exist } c \in (a, b) \text{ for which } \frac{f'''(c)}{f(c)} \text{ is}$$

Sol. Let $\phi(x) = (f(x) + f'(x) + f''(x))e^{-x}$

$\Rightarrow \phi(x)$ is differential on (a, b) and continuous on $[a, b]$.

$$\begin{aligned} \text{Also, } \frac{\phi(a)}{\phi(b)} &= \frac{f(a) + f'(a) + f''(a)}{f(b) + f'(b) + f''(b)} \times e^{b-a} \\ &= e^{a-b} \times e^{b-a} = e^{a-b+b-a} = e^0 = 1 \end{aligned}$$

$$\Rightarrow \phi(a) = \phi(b)$$

Thus, Rolle's theorem is satisfied

Now, there exists a point $c \in (a, b)$ such that

$$\phi'(c) = 0$$

$$\Rightarrow (f'(c) + f''(c) + f'''(c))e^{-c} - (f(c) + f'(c) + f''(c))e^{-c} = 0$$

$$\Rightarrow f'(c) + f''(c) + f'''(c) - f(c) - f'(c) - f''(c) = 0$$

$$\Rightarrow f'''(c) = f(c)$$

$$\Rightarrow \frac{f'''(c)}{f(c)} = 1$$

16. A function $f: R \rightarrow R$ is defined as $f(x) = |x|^m |x-1|^n \forall x \in (0, 1), m, n \in N$. Then the maximum value of the function is _____

Sol. $\therefore f(x) = \begin{cases} (-1)^{m+n} \times x^m (x-1)^n & , \quad x < 0 \\ (-1)^n \times x^m \times (x-1)^n & , \quad 0 \leq x < 1 \\ x^m (x-1)^n & , \quad x \geq 1 \end{cases}$

Let $g(x) = x^m (x-1)^n$

$$g'(x) = mx^{m-1}(x-1)^n + nx^m(x-1)^{n-1}$$

$$= x^{m-1}(x-1)^{n-1} \{mx - m + nx\} = 0$$

Now, $f'(x) = 0$

$\Rightarrow g'(x) = 0$

$\Rightarrow x = 0, \frac{m}{m+n}$

Now, $f(0) = 0, f(1) = 1$

And $f\left(\frac{m}{m+n}\right) = (-1)^n \times \frac{m^m n^n (-1)^n}{(m+n)^{m+n}} = \frac{m^m n^n}{(m+n)^{m+n}} > 1$

Hence, the maximum value = $\frac{m^m n^n}{(m+n)^{m+n}}$

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